

Hemodynamic Disorders, Thromboembolic Disease, and Shock

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The health of cells and tissues depends on the circulation of blood, which delivers oxygen and nutrients and removes wastes generated by cellular metabolism. Under normal conditions, as blood passes through capillary beds, proteins in the plasma are retained within the vasculature, and there is little net movement of water and electrolytes into the tissues. This balance is often disturbed by pathologic conditions that alter endothelial function, increase vascular hydrostatic pressure, or decrease plasma protein content, all of which promote edema—the accumulation of fluid in tissues resulting from a net movement of water into extravascular spaces. Depending on its severity and location, edema may have minimal or profound effects. In the lower extremities, it may only make one's shoes feel snugger after a long sedentary day; in the lungs, however, edema fluid can fill alveoli, causing life-threatening hypoxia.

The structural integrity of blood vessels is frequently compromised by trauma or intrinsic disorders. *Hemostasis* is the process of blood clotting that prevents excessive bleeding after blood vessel damage. Inadequate hemostasis may result in hemorrhage, which can compromise regional tissue perfusion and, if massive and rapid, may lead to hypotension, shock, and death. Conversely, inappropriate clotting (*thrombosis*) or migration of clots (*embolism*) can obstruct blood vessels, potentially causing ischemic cell death (*infarction*). Indeed, thromboembolism lies at the heart of three major causes of morbidity and death in high income countries: myocardial infarction, pulmonary embolism (PE), and cerebrovascular accident (stroke).

Herein, we focus on disorders of hemodynamics (edema, effusions, congestion, and shock), provide an overview of disorders of abnormal bleeding and clotting (thrombosis), and discuss the various forms of embolism.

EDEMA AND EFFUSIONS

Disorders that perturb cardiovascular, renal, or hepatic function are often marked by the accumulation of fluid in tissues (edema) or body cavities (effusions). Under normal circumstances, the tendency

of vascular hydrostatic pressure to push water and salts out of capillaries into the interstitial space is nearly balanced by the tendency of plasma colloid osmotic pressure to pull water and salts back into vessels. There is usually a small net movement of fluid into the interstitium, but this drains into lymphatic vessels and ultimately returns to the bloodstream via the thoracic duct, keeping the tissues “dry” (Fig. 4.1). **Elevated hydrostatic pressure or diminished colloid osmotic pressure disrupts this balance and results in increased movement of fluid out of vessels.** If the net rate of fluid influx exceeds the rate of lymphatic drainage, fluid accumulates. Within tissues the result is edema, and if a serosal surface is involved, fluid may accumulate within the adjacent body cavity as an effusion.

Edema fluids and effusions may be *inflammatory* or *noninflammatory* (Table 4.1). Inflammation-related edema and effusions are discussed in detail in Chapter 3. These protein-rich *exudates* accumulate due to increases in vascular permeability caused by inflammatory mediators. In contrast, noninflammatory edema and effusions are protein-poor fluids called *transudates*. Noninflammatory edema and effusions are common in many disorders, including heart failure, liver failure, renal disease, and malnutrition (Fig. 4.2). We will now discuss the various causes of edema.

Increased Hydrostatic Pressure

Increases in hydrostatic pressure are mainly caused by disorders that impair venous return. If the impairment is localized (e.g., a deep venous thrombosis [DVT] in a lower extremity), then the resulting edema is confined to the affected part. Conditions leading to systemic increases in venous pressure (e.g., congestive heart failure, Chapter 12) are understandably associated with more widespread edema.

Reduced Plasma Osmotic Pressure

Under normal circumstances albumin accounts for almost half of the total plasma protein; it follows that conditions leading to inadequate

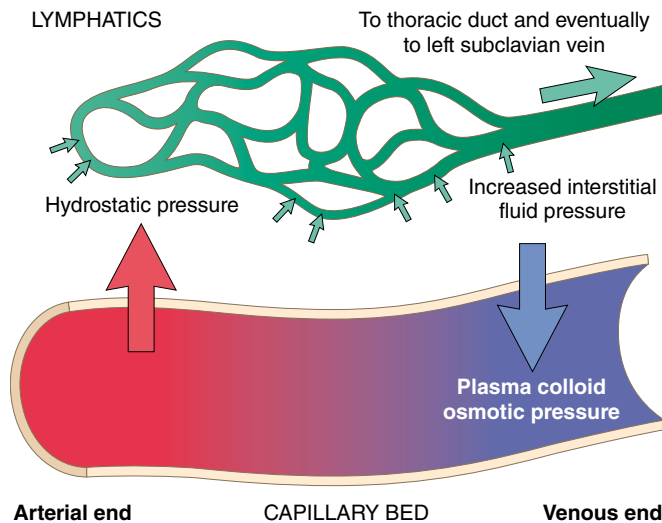


Fig. 4.1 Factors influencing fluid movement across capillary walls. Normally, hydrostatic and osmotic forces are nearly balanced so that there is little net movement of fluid out of vessels. Many different pathologic disorders (see Table 4.1) are associated with increases in capillary hydrostatic pressure or decreases in plasma osmotic pressure that lead to the extravasation of fluid into tissues. Lymphatic vessels remove much of the excess fluid, but if the capacity for lymphatic drainage is exceeded, tissue edema results.

TABLE 4.1 Pathophysiologic Categories of Edema

Increased Hydrostatic Pressure

Impaired Venous Return

Congestive heart failure
Constrictive pericarditis
Ascites (liver cirrhosis)
Venous obstruction or compression
• Thrombosis
• External pressure (e.g., mass)
Lower extremity inactivity with prolonged dependency

Arteriolar Dilation

Heat
Neurohumoral dysregulation

Reduced Plasma Osmotic Pressure (Hypoproteinemia)

Protein-losing glomerulopathies (nephrotic syndrome)
Liver cirrhosis
Malnutrition
Protein-losing gastroenteropathy

Lymphatic Obstruction

Inflammatory
Neoplastic
Postsurgical
Postirradiation

Sodium Retention

Excessive salt intake with renal insufficiency
Increased tubular reabsorption of sodium
Renal hypoperfusion
Increased renin-angiotensin-aldosterone secretion

Inflammation

Acute inflammation
Chronic inflammation
Angiogenesis

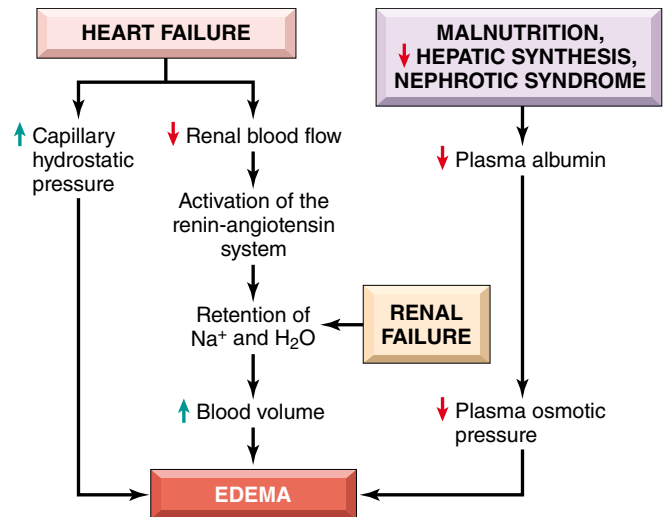


Fig. 4.2 Mechanisms of systemic edema in heart failure, renal failure, malnutrition, hepatic failure, and nephrotic syndrome.

synthesis or increased loss of albumin from the circulation are common causes of reduced plasma oncotic pressure. Reduced albumin synthesis occurs mainly in severe liver diseases (e.g., end-stage cirrhosis, Chapter 18) and protein malnutrition (Chapter 9). An important cause of albumin loss is the *nephrotic syndrome* (Chapter 20), in which albumin leaks into the urine through abnormally permeable glomerular capillaries. Regardless of cause, reduced plasma osmotic pressure leads to edema caused by increased movement of fluid out of vessels; this in turn results in reduced intravascular volume, renal hypoperfusion, and secondary hyperaldosteronism. Not only does the ensuing salt and water retention by the kidney fail to correct the plasma volume deficit, but it also exacerbates the edema, because the primary defect—a low plasma protein level—persists.

Sodium and Water Retention

Increased salt retention—with obligate retention of associated water—causes both increased hydrostatic pressure (due to intravascular fluid volume expansion) and diminished vascular colloid osmotic pressure (due to dilution). Salt retention occurs whenever renal function is compromised, such as in primary kidney disorders and in cardiovascular disorders that decrease renal perfusion. One of the most important causes of renal hypoperfusion is congestive heart failure, which (like hypoproteinemia) results in the activation of the *renin-angiotensin-aldosterone axis*. In early heart failure, this response is beneficial, as the retention of sodium and water and other adaptations, including increased vascular tone and elevated levels of antidiuretic hormone, improve cardiac output and restore normal renal perfusion. However, as heart failure worsens and cardiac output diminishes, the retained fluid merely increases the hydrostatic pressure, leading to edema and effusions.

Lymphatic Obstruction

Trauma, fibrosis, invasive tumors, and infectious agents can all disrupt lymphatic vessels and impair the clearance of interstitial fluid, resulting in lymphedema in the affected part of the body. A dramatic example is seen in parasitic *filariasis*, in which the organism induces obstructive fibrosis of lymphatic channels and lymph nodes. This may result in edema of the external genitalia and lower limbs that is so massive as to earn the appellation *elephantiasis*. Severe edema of the upper extremity may also complicate surgical removal and/or irradiation of the breast and associated axillary lymph nodes in patients with breast cancer.

MORPHOLOGY

Edema is easily recognized grossly; microscopically, it is appreciated as clearing and separation of the extracellular matrix (ECM) and subtle cell swelling. Edema is most commonly seen in subcutaneous tissues, the lungs, and the brain. **Subcutaneous edema** can be diffuse or more conspicuous in regions with high hydrostatic pressures. Its distribution is often influenced by gravity (e.g., it appears in the legs when standing and the sacrum when recumbent), a feature termed **dependent edema**. Finger pressure over markedly edematous subcutaneous tissue displaces the interstitial fluid and leaves a depression, a sign called **pitting edema**.

Edema resulting from renal dysfunction often appears initially in parts of the body containing loose connective tissue, such as the eyelids; **periorbital edema** is thus a characteristic finding in severe renal disease. With **pulmonary edema**, the lungs are often two to three times their normal weight, and sectioning yields frothy, blood-tinged fluid—a mixture of air, edema, and extravasated red cells. **Brain edema** can be localized or generalized depending on the nature and extent of the pathologic process or injury. The swollen brain exhibits narrowed sulci and distended gyri, which are compressed by the unyielding skull (Chapter 28).

Effusions involving the pleural cavity (**hydrothorax**), the pericardial cavity (**hydropericardium**), or the peritoneal cavity (**hydroperitoneum** or **ascites**) are common in a wide range of clinical settings. Transudative effusions are typically protein poor, translucent, and straw colored; an exception is peritoneal effusion caused by lymphatic blockage (**chylous effusion**), which may be milky due to the presence of lipids absorbed from the gut. In contrast, exudative effusions are protein-rich and often cloudy due to the presence of white cells.

Clinical Features

The consequences of edema range from merely annoying to rapidly fatal. Subcutaneous edema is important primarily because it signals potential underlying cardiac or renal disease; however, when significant, it can also impair wound healing and the clearance of infections. Pulmonary edema is a common clinical problem that is most frequently seen in the setting of left ventricular failure; it can also occur with renal failure, acute respiratory distress syndrome (Chapter 15), and pulmonary inflammation or infection. Edema in the pulmonary interstitium and the alveolar spaces impedes gas exchange (leading to hypoxemia) and creates a favorable environment for bacterial infection. Pulmonary edema is often exacerbated by pleural effusions, which may further compromise gas exchange by compressing the underlying pulmonary parenchyma. Peritoneal effusions (ascites) resulting most commonly from portal hypertension are prone to seeding by bacteria, leading to serious and sometimes fatal infections. Brain edema is life threatening; if severe, brain substance can herniate (extrude) through the foramen magnum, or the brainstem vascular supply can be compressed. Either condition can injure the medullary centers and cause death (Chapter 28).

KEY CONCEPTS

Edema

Edema is the result of the movement of fluid from the vasculature into the interstitial spaces; the fluid may be protein poor (transudate) or protein rich (exudate). Edema may be caused by:

- Increased hydrostatic pressure (e.g., heart failure)
- Decreased colloid osmotic pressure caused by reduced plasma albumin, either due to decreased synthesis (e.g., liver disease, protein malnutrition) or to increased loss (e.g., nephrotic syndrome)
- Increased vascular permeability (e.g., inflammation)
- Lymphatic obstruction (e.g., infection or neoplasia)
- Sodium and water retention (e.g., renal failure)

HYPEREMIA AND CONGESTION

Hyperemia and congestion both stem from increased blood volumes within tissues, but have different underlying mechanisms and consequences. *Hyperemia* is an active process in which arteriolar dilation (e.g., at sites of inflammation or in skeletal muscle during exercise) leads to increased blood flow. Affected tissues turn red (*erythema*) because of increased delivery of oxygenated blood. *Congestion* is a passive process resulting from reduced venous outflow of blood from a tissue. It can be systemic, as in cardiac failure, or localized, as in isolated venous obstruction. Congested tissues have an abnormal blue-red color (*cyanosis*) that stems from the accumulation of deoxygenated hemoglobin in the affected area. In long-standing *chronic passive congestion*, the associated chronic hypoxia may result in ischemic tissue injury and scarring. In chronically congested tissues, capillary rupture can also produce small hemorrhagic foci; subsequent catabolism of extravasated red cells can leave residual telltale clusters of hemosiderin-laden macrophages. As a result of increased hydrostatic pressures, congestion commonly leads to edema.

MORPHOLOGY

Congested tissues take on a dusky reddish-blue color (*cyanosis*) due to red cell stasis and the presence of deoxygenated hemoglobin. Microscopically, **acute pulmonary congestion** is marked by engorged alveolar capillaries, alveolar septal edema, and focal intraalveolar hemorrhage. In **chronic pulmonary congestion**, which is often caused by congestive heart failure, the septa are thickened and fibrotic, and the alveoli often contain numerous macrophages laden with hemosiderin (**heart failure cells**) derived from phagocytosed red cells. In **acute hepatic congestion**, the central vein and sinusoids are distended. Because the centrilobular area is at the distal end of the hepatic blood supply, centrilobular hepatocytes may undergo ischemic necrosis, and the periportal hepatocytes—better oxygenated because of proximity to hepatic arterioles—may only develop fatty change. In **chronic passive hepatic congestion**, the centrilobular regions are grossly red-brown and slightly depressed (because of cell death) and are accentuated against the surrounding zones of uncongested tan liver (**nutmeg liver**) (Fig. 4.3A). Microscopically, there is centrilobular congestion and hemorrhage, hemosiderin-laden macrophages, and variable degrees of hepatocyte dropout and necrosis (see Fig. 4.3B).

HEMOSTASIS, HEMORRHAGIC DISORDERS, AND THROMBOSIS

Hemostasis can be defined simply as the process by which blood clots form at sites of vascular injury. Hemostasis is essential for life and is deranged to varying degrees in a broad range of disorders, which can be divided into two groups. In hemorrhagic disorders, characterized by excessive bleeding, hemostatic mechanisms are either blunted or insufficient to prevent blood loss. By contrast, in thrombotic disorders, blood clots (often referred to as thrombi) form within intact blood vessels or within the chambers of the heart. As is discussed in Chapters 11 and 12, thrombosis has a central role in the most common and clinically important forms of cardiovascular disease.

Although useful, it must be recognized that this division between bleeding and thrombotic disorders sometimes breaks down, in that generalized activation of clotting sometimes paradoxically produces bleeding due to the consumption of coagulation factors, as in *disseminated intravascular coagulation* (DIC). To provide context for

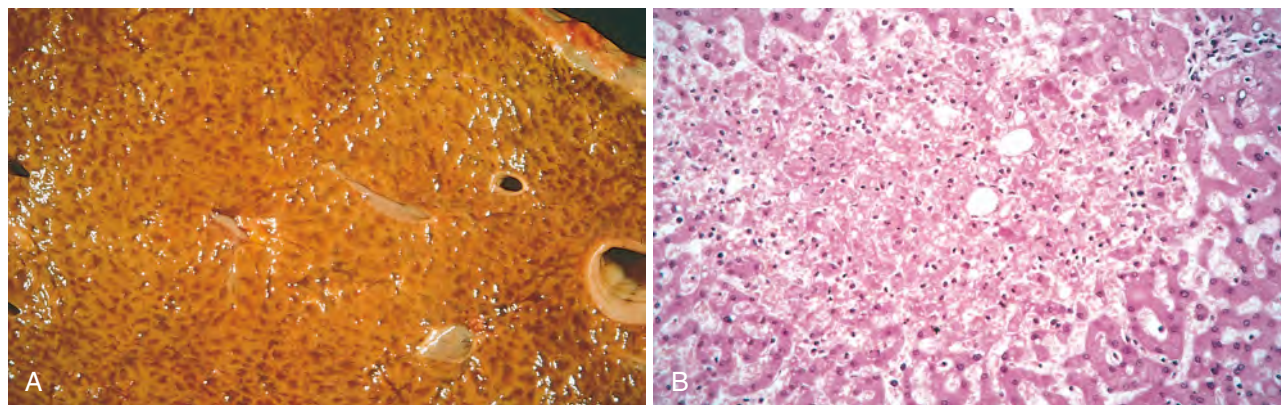


Fig. 4.3 Liver with chronic passive congestion and hemorrhagic necrosis. (A) Central areas are red and slightly depressed compared with the surrounding tan viable parenchyma, forming a “nutmeg liver” pattern (so-called because it resembles the cut surface of a nutmeg). (B) Centrilobular necrosis with degenerating hepatocytes and hemorrhage. (Courtesy Dr. James Crawford, Department of Pathology, University of Florida, Gainesville, FL.)

understanding disorders of bleeding and clotting, this discussion begins with normal hemostasis, focusing on the contribution of platelets, coagulation factors, and endothelium.

Normal Hemostasis

Hemostasis is a precisely orchestrated process involving platelets, clotting factors, and endothelium that occurs at the site of vascular injury and culminates in the formation of a blood clot, which serves to prevent or limit the extent of bleeding. The general sequence of events leading to hemostasis at a site of vascular injury is shown in Fig. 4.4.

- **Vasoconstriction:** *arteriolar vasoconstriction* occurs immediately and markedly reduces blood flow to the injured area (see Fig. 4.4A). It is mediated by reflex neurogenic mechanisms and may be augmented by the local secretion of factors such as *endothelin*, a potent endothelium-derived vasoconstrictor. This effect is transient, however, and bleeding would resume if not for activation of platelets and coagulation factors.
- **Primary hemostasis:** *the formation of the platelet plug.* Disruption of the endothelium exposes subendothelial *von Willebrand factor* (vWF) and collagen, which promote platelet adherence and activation. Activation of platelets results in a dramatic shape change from small, rounded disks to flat plates with spiky protrusions that markedly increase surface area, as well as the release of secretory granules. Within a few minutes, the secreted products recruit additional platelets that undergo aggregation to form a primary hemostatic plug (see Fig. 4.4B).
- **Secondary hemostasis:** *deposition of fibrin.* Vascular injury exposes *tissue factor* (TF) at the site of injury. Tissue factor is a membrane-bound procoagulant glycoprotein that is normally expressed by subendothelial cells in the vessel wall, such as smooth muscle cells and fibroblasts. Tissue factor binds and activates factor VII (see later), setting in motion a cascade of reactions that culminates in *thrombin* generation. Thrombin cleaves circulating fibrinogen into insoluble *fibrin*, creating a fibrin meshwork, and is a potent activator of platelets, leading to additional platelet aggregation at the site of injury. This sequence, referred to as *secondary hemostasis*, consolidates the initial platelet plug (see Fig. 4.4C).
- **Clot stabilization and resorption.** Polymerized fibrin and platelet aggregates undergo contraction to form a solid, permanent plug that prevents further hemorrhage. At this stage, counterregulatory

mechanisms (e.g., tissue plasminogen activator [t-PA] made by endothelial cells) are set into motion that limit clotting to the site of injury (see Fig. 4.4D) and eventually lead to clot resorption and tissue repair.

It should be emphasized that endothelial cells are central regulators of hemostasis; the balance between the antithrombotic and prothrombotic activities of endothelium determines whether thrombus formation, propagation, or dissolution occur. Normal endothelial cells express a variety of *anticoagulant* factors that inhibit platelet aggregation and coagulation and promote fibrinolysis; after injury or activation, however, this balance shifts, and endothelial cells acquire numerous *procoagulant* activities (activation of platelets and clotting factor, described earlier; see also Fig. 4.10). Besides trauma, endothelium can be activated by microbial pathogens, hemodynamic forces, and proinflammatory mediators.

We now describe roles of platelets, coagulation factors, and endothelium in hemostasis in greater detail, following the scheme illustrated in Fig. 4.4.

Platelets

Platelets play a critical role in hemostasis by forming the primary plug that initially seals vascular defects and by providing a surface that binds and concentrates activated coagulation factors. Platelets are disk-shaped anucleate cell fragments that are shed from megakaryocytes in the bone marrow into the bloodstream. Their function depends on several glycoprotein receptors, a contractile cytoskeleton, and two types of cytoplasmic granules. *α-Granules* have the adhesion molecule P-selectin on their membranes (Chapter 3) and contain proteins involved in coagulation, such as fibrinogen; coagulation factors V, XI, and vWF; as well as protein factors that may be involved in wound healing, such as fibronectin, platelet factor 4 (PF4, a heparin-binding chemokine), platelet-derived growth factor (PDGF), and transforming growth factor-β (TGF-β). *Dense* (or *δ*) *granules* contain adenosine diphosphate (ADP), adenosine triphosphate (ATP), ionized calcium, serotonin, and epinephrine. Platelet membranes express several glycoprotein (Gp) receptors that play an important role in hemostasis as described below. They include (with their ligands in parentheses) Gp Ib (vWF); Gp VI and Gp Ia/IIa complex (collagen); and Gp IIb/IIIa complex (fibrinogen).

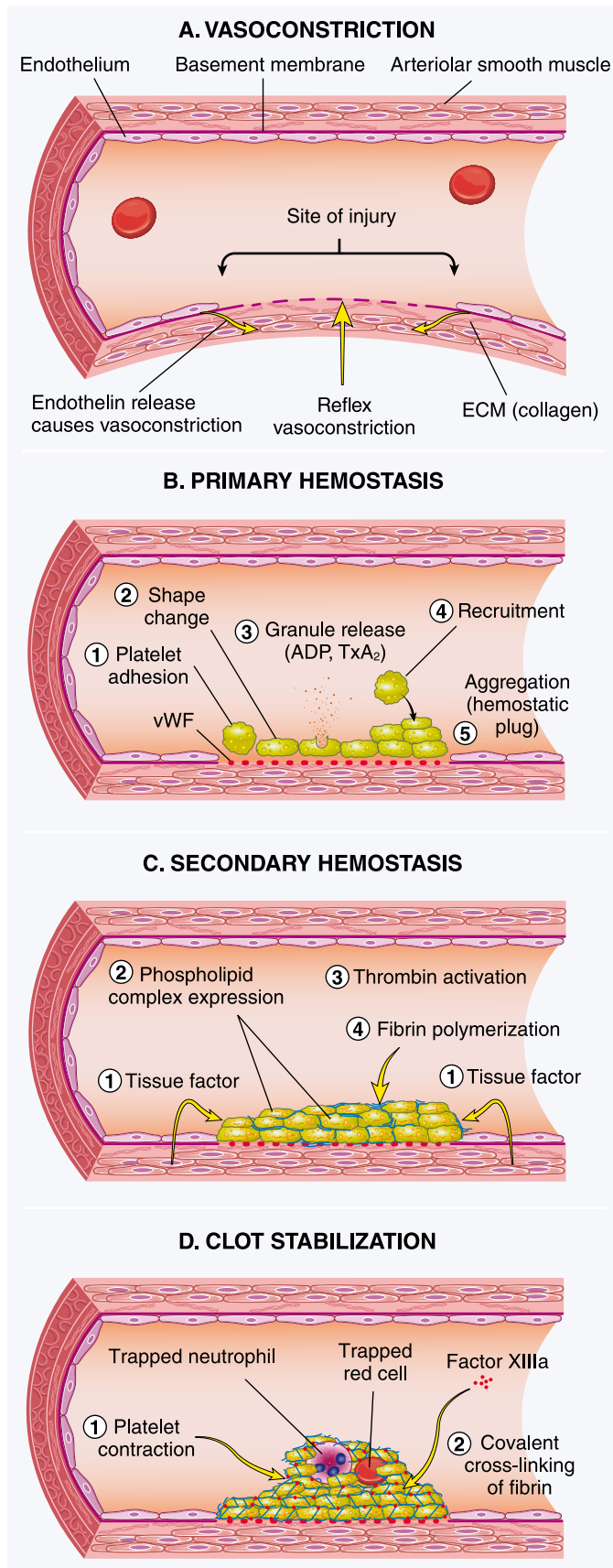


Fig. 4.4 Normal hemostasis. (A) After vascular injury, neurohumoral factors induce transient vasoconstriction. (B) Platelets bind via glycoprotein Ib (GpIb) receptors to von Willebrand factor (vWF) on exposed extracellular matrix (ECM) and are activated, undergoing a shape change and granule release. Released adenosine diphosphate (ADP) and thromboxane A_2 (TxA_2) induce additional platelet aggregation through platelet GpIIb/IIIa receptor binding to fibrinogen, and form the primary hemostatic plug. (C) Local activation of the coagulation cascade (involving tissue factor and platelet phospholipids) results in fibrin polymerization, “cementing” the platelets into a definitive secondary hemostatic plug. (D) Counterregulatory mechanisms, mediated by tissue plasminogen activator (t-PA, a fibrinolytic product) and thrombomodulin, confine the hemostatic process to the site of injury.

After a traumatic vascular injury, platelets encounter constituents of the subendothelial connective tissue, such as vWF and collagen. On contact with these proteins, platelets undergo a sequence of reactions that culminate in the formation of a platelet plug (see Fig. 4.4B).

- **Platelet adhesion** is mediated largely via interactions between GpIb and vWF in the subendothelial matrix (Fig. 4.5). Under high shear stresses, vWF can bind both to platelets and to collagen, thus acting like a glue between them. Platelets also adhere to exposed collagen via the platelet collagen receptors GpIa/IIa and GpVI. Notably, genetic deficiencies of vWF (*von Willebrand disease*, Chapter 14) or GpIb (*Bernard-Soulier syndrome*) result in bleeding disorders, attesting to the importance of these factors.
- **Platelets rapidly change shape** following adhesion, being converted from smooth disks to spiky “sea urchins” with greatly increased surface area. This change is accompanied by conformational changes in cell surface GpIIb/IIIa that increase its affinity for fibrinogen and by the translocation of *negatively charged phospholipids* (particularly phosphatidylserine) to the platelet surface. These phospholipids bind calcium and serve as nucleation sites for the assembly of coagulation factor complexes.
- **Secretion (release reaction) of granule contents** occurs along with changes in shape; these two events are often referred to together as platelet activation. Platelet activation is triggered by a number of factors, including the coagulation factor thrombin and ADP. Thrombin activates platelets through a special G protein–coupled receptor referred to as *protease-activated receptor-1* (PAR-1), which is switched on by a proteolytic cleavage carried out by thrombin. ADP is a component of dense-body granules; thus, platelet activation and ADP release begets additional rounds of platelet activation, a phenomenon referred to as *recruitment*. ADP acts by binding two G protein–coupled receptors (P2Y_1 and P2Y_{12}). Activated platelets also produce the prostaglandin *thromboxane* A_2 (TxA_2), a potent inducer of platelet aggregation. An understanding of the biochemical pathways involved in platelet activation has led to the development of drugs with antiplatelet activities. *Aspirin*, the oldest drug of all, inhibits platelet aggregation and produces a mild bleeding defect by inhibiting cyclooxygenase, a platelet enzyme that is required for TxA_2 synthesis. More recently, drugs that inhibit platelet function by antagonizing protease-activated receptor 1 or P2Y_{12} have been developed for the treatment of coronary artery disease.
- **Platelet aggregation** follows their activation. The conformational change in GpIIb/IIIa that occurs with platelet activation allows binding of fibrinogen, a large bivalent plasma polypeptide that forms bridges between adjacent platelets, leading to their aggregation. Predictably, inherited deficiency of GpIIb/IIIa results in a bleeding disorder, which is called *Glanzmann thrombasthenia*. The initial wave of aggregation is reversible, but concurrent activation

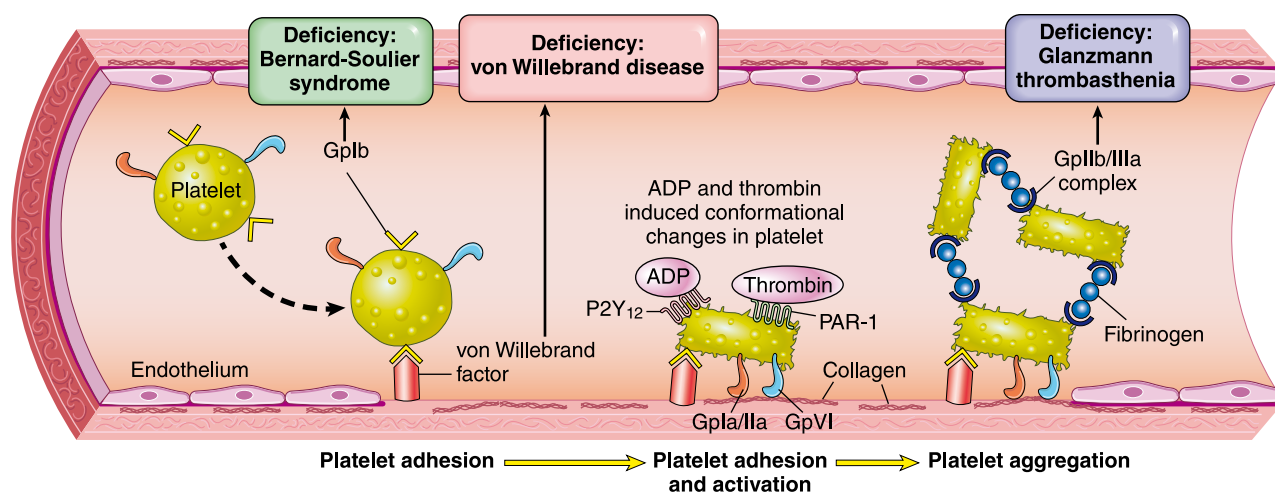


Fig. 4.5 Platelet adhesion and aggregation. von Willebrand factor functions as an adhesion bridge between subendothelial collagen and the glycoprotein Ib (*GpIb*) platelet receptor. Platelets also adhere to exposed collagen via the platelet collagen receptors *GpIa/IIa* and *GpVI*. Aggregation is accomplished by fibrinogen bridging *GpIIb/IIIa* receptors on different platelets. Congenital deficiencies in the various receptors or bridging molecules lead to bleeding disorders indicated in the colored boxes. *ADP*, Adenosine diphosphate; *PAR-1*, protease-activated receptor 1.

of thrombin stabilizes the platelet plug by causing further platelet activation and aggregation, and by promoting irreversible platelet contraction. Platelet contraction is dependent on the cytoskeleton and consolidates the aggregated platelets. In parallel, thrombin also converts fibrinogen into insoluble *fibrin*, cementing the platelets in place and creating the definitive *secondary hemostatic plug*. Entrapped red cells and leukocytes are also found in hemostatic plugs, in part due to adherence of leukocytes to P-selectin expressed on activated platelets.

KEY CONCEPTS

Platelet Adhesion, Activation, and Aggregation

- Endothelial injury exposes the underlying basement membrane ECM; platelets adhere to the ECM primarily through the binding of platelet *GpIb* receptors to vWF, which in turn binds to subendothelial collagen.
- Adhesion leads to platelet activation, an event associated with secretion of platelet granule contents, including calcium (a cofactor for several coagulation proteins) and ADP (a mediator of further platelet activation); dramatic changes in shape and membrane composition; and activation of *GpIIb/IIIa* receptors.
- The *GpIIb/IIIa* receptors on activated platelets form bridging cross-links with fibrinogen, leading to platelet aggregation.
- Concomitant activation of thrombin promotes fibrin deposition, cementing the platelet plug in place.

Coagulation Cascade

The **coagulation cascade** is a series of amplifying enzymatic reactions that lead to the deposition of an insoluble fibrin clot. The principal mediators of coagulation are a set of plasma proteins that circulate in an inactive form and as described below are activated during clot formation. An exception is tissue factor, which is constitutively expressed on cells within blood vessels and is exposed only after tissue injury. As discussed later, the dependency of clot formation on various factors differs in the laboratory test tube and in blood vessels in vivo (Fig. 4.6). However, clotting in vitro and in vivo both follow the same general principles.

The cascade of reactions in the pathway can be likened to a “dance” in which coagulation factors are passed from one partner to the next (Fig. 4.7). Each reaction step involves an *enzyme* (an activated coagulation factor), a *substrate* (an inactive proenzyme form of a coagulation factor), and a *cofactor* (a reaction accelerator). These components are assembled on a negatively charged phospholipid surface, which is provided by activated platelets. Assembly of reaction complexes also depends on calcium, which binds to γ -carboxylated glutamic acid residues that are present in factors II, VII, IX, and X. The enzymatic reactions that produce γ -carboxylated glutamic acid use vitamin K as a cofactor and are antagonized by drugs such as coumadin, used as an anticoagulant.

Based on assays performed in clinical laboratories, the coagulation cascade can be divided into the *extrinsic* and *intrinsic* pathways (see Fig. 4.6A).

- The *prothrombin time* (PT) assay assesses the function of the proteins in the extrinsic pathway (factors VII, X, V, II [prothrombin], and fibrinogen). In brief, tissue factor, phospholipids, and calcium are added to plasma, and the time for a fibrin clot to form is recorded.
- The *partial thromboplastin time* (PTT) assesses the function of the proteins in the intrinsic pathway (factors XII, XI, IX, VIII, X, V, II, and fibrinogen). In this assay, clotting of plasma is initiated by the addition of negatively charged particles (e.g., ground glass) that activate factor XII (Hageman factor) together with phospholipids and calcium, and the time to fibrin clot formation is recorded.

Although the PT and PTT assays are of great utility in evaluating coagulation factor function in patients, they do not recapitulate the events that lead to coagulation in vivo. This point is most clearly made by considering the clinical effects of deficiencies of various coagulation factors. Deficiencies of factors V, VII, VIII, IX, and X are associated with moderate to severe bleeding disorders, and prothrombin deficiency is likely incompatible with life. By contrast, factor XI deficiency is only associated with mild bleeding, suggesting that factor XI and the intrinsic pathway play a minor role in hemostasis, and in fact there is accumulating evidence that factor XI is involved in pathologic thrombosis.

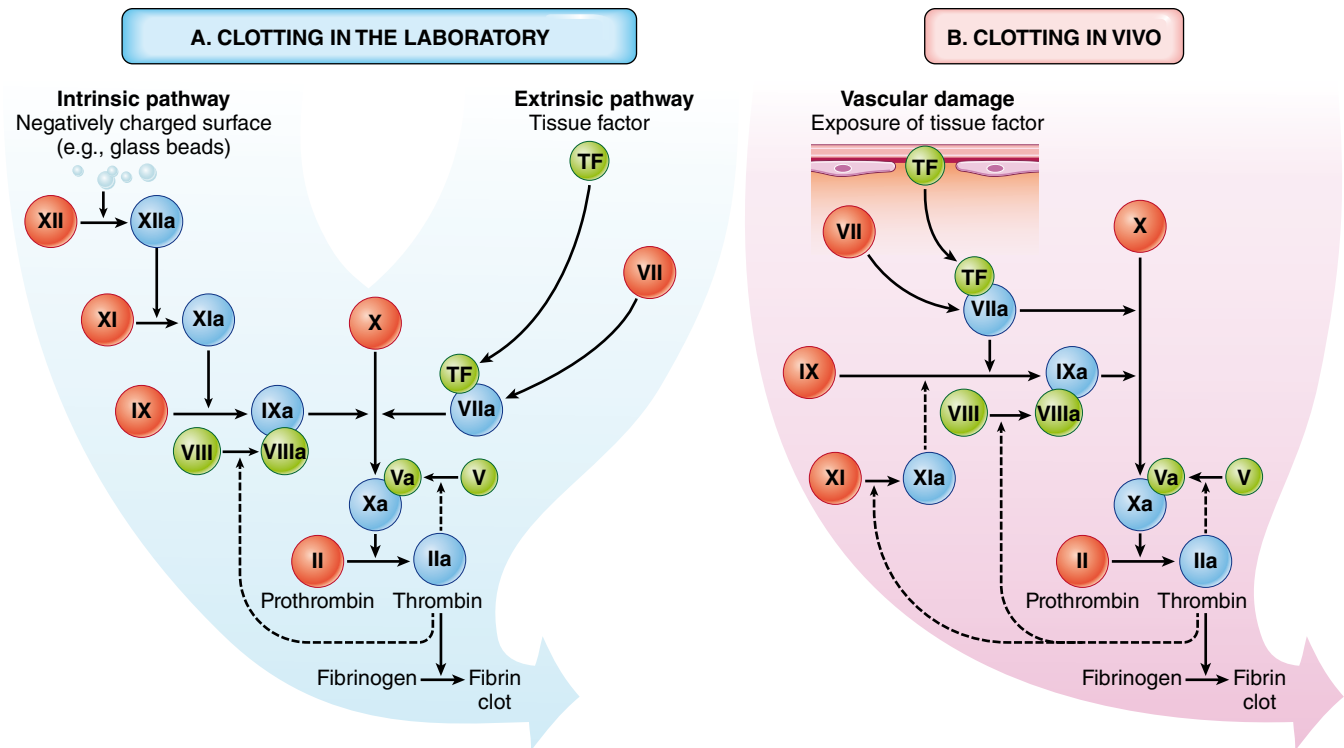


Fig. 4.6 The coagulation cascade in the laboratory and in vivo. (A) Clotting is initiated in the laboratory by adding phospholipids, calcium, and either a negatively charged substance such as glass beads (intrinsic pathway) or a source of tissue factor (extrinsic pathway). (B) In vivo, tissue factor is the major initiator of coagulation, which is amplified by feedback loops involving thrombin (dotted lines). The red polypeptides are inactive factors, the blue polypeptides are active factors, and the green polypeptides correspond to cofactors (reaction accelerators).

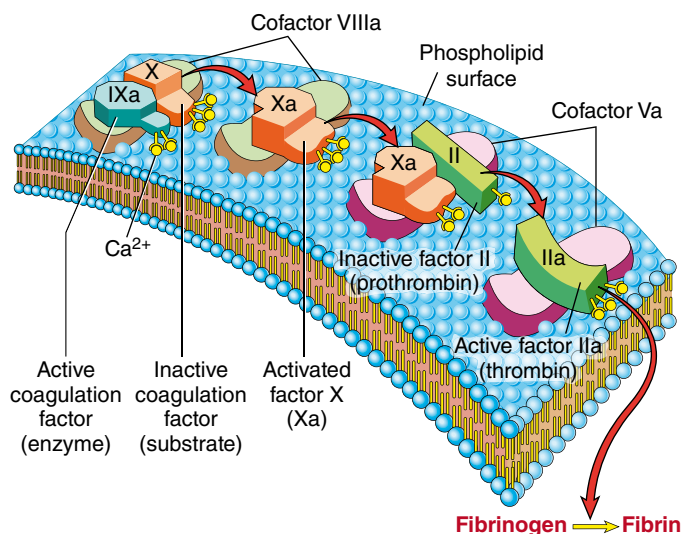


Fig. 4.7 Conversion of factor X to factor Xa, which in turn converts factor II (prothrombin) to factor IIa (thrombin) (see Fig. 4.6B). The initial reaction complex consists of a proteolytic enzyme (factor IXa), a substrate (factor X), and a reaction accelerator (factor VIIIa), all assembled on a platelet phospholipid surface. Calcium ions hold the assembled components together and are essential for the reaction. Activated factor Xa becomes the protease for the second adjacent complex in the coagulation cascade, converting prothrombin substrate (II) to thrombin (IIa) using factor Va as the reaction accelerator.

Based on the effects of various factor deficiencies in humans, it is believed that, in vivo, factor VIIa/tissue factor complex is the most important activator of factor IX and that factor IXa/factor VIIIa complex is the most important activator of factor X. Of note, factor VIIa–tissue factor activates factor X directly as well as via activation of factor IX (see Fig 4.6B). The mild bleeding tendency seen in patients with factor XI deficiency is likely explained by the ability of thrombin to activate factor XI (as well as factors V and VIII), a feedback mechanism that amplifies the coagulation cascade.

Among the coagulation factors, thrombin is the most important, because its various enzymatic activities control diverse aspects of hemostasis and link clotting to inflammation and repair. Among thrombin's most important activities are:

- **Conversion of fibrinogen into cross-linked fibrin.** Thrombin directly converts soluble fibrinogen into fibrin monomers that polymerize into an insoluble fibril, and amplifies the coagulation process, not only by activating factor X and XI, but also by activating two critical cofactors: factors V and VIII. It also stabilizes the secondary hemostatic plug by activating factor XIII, which covalently cross-links fibrin.
- **Platelet activation.** Thrombin is a potent inducer of platelet activation and aggregation through its ability to activate PAR-1, thereby linking platelet function to coagulation.
- **Proinflammatory effects.** PARs also are expressed on inflammatory cells, endothelium, and other cell types (Fig. 4.8), and activation of these receptors by thrombin is believed to mediate proinflammatory effects that contribute to tissue repair and angiogenesis.

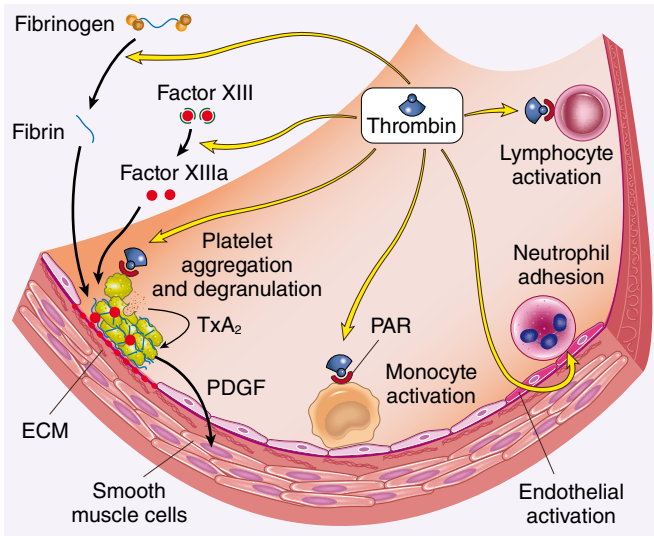


Fig. 4.8 Role of thrombin in hemostasis and cellular activation. Thrombin plays a critical role in generating cross-linked fibrin (by cleaving fibrinogen to fibrin and by activating factor XIII), as well as activating several other coagulation factors (see Fig. 4.6B). Through protease-activated receptors (PAR, see text), thrombin also modulates several cellular activities. It directly induces platelet aggregation and thromboxane A_2 (TxA_2) production, and activates endothelial cells, which respond by expressing adhesion molecules and cytokine mediators (e.g., $PDGF$). Thrombin also directly activates leukocytes. *ECM*, Extracellular matrix; $PDGF$, platelet-derived growth factor. See Fig. 4.10 for additional anticoagulant activities mediated by thrombin. (Courtesy Shaun Coughlin, MD, PhD, Cardiovascular Research Institute, University of California San Francisco.)

- **Anticoagulant effects.** Remarkably, through mechanisms described later, on encountering normal endothelium, thrombin changes from a procoagulant to an anticoagulant; this reversal in function prevents clots from extending beyond the site of the vascular injury.

Factors That Limit Coagulation. Once initiated, coagulation must be restricted to the site of vascular injury to prevent deleterious consequences. One limiting factor is simple dilution; blood flowing past the site of injury washes out activated coagulation factors, which are rapidly removed by the liver. A second is the requirement for negatively charged phospholipids, which, as mentioned, are mainly provided by

platelets that have been activated by contact with subendothelial matrix at sites of vascular injury. However, the most important counter-regulatory mechanisms involve factors that are expressed by intact endothelium adjacent to the site of injury (described later).

Activation of the coagulation cascade also sets into motion a fibrinolytic cascade that limits the size of the clot and contributes to its later dissolution (Fig. 4.9). Fibrinolysis is largely accomplished through the enzymatic activity of *plasmin*, which breaks down fibrin and interferes with its polymerization. An elevated level of breakdown products of fibrinogen (often called fibrin split products), most notably fibrin-derived *D-dimers*, are a useful clinical marker of several thrombotic states (described later). Plasmin is generated by enzymatic catabolism of the inactive circulating precursor *plasminogen*, either by a factor XII-dependent pathway (possibly explaining the association of factor XII deficiency and thrombosis) or by plasminogen activators. The most important plasminogen activator is t-PA; it is synthesized principally by endothelium and is most active when bound to fibrin. This characteristic makes t-PA a useful therapeutic agent, since its fibrinolytic activity is largely confined to sites of recent thrombosis. Once activated, plasmin is in turn tightly controlled by counterregulatory factors such as α_2 -plasmin inhibitor, a plasma protein that binds and rapidly inhibits free plasmin.

Endothelium

The balance between the anticoagulant and procoagulant activities of endothelium often determines whether clot formation, propagation, or dissolution occurs (Fig. 4.10). Normal endothelial cells express a multitude of factors that inhibit the procoagulant activities of platelets and coagulation factors and that augment fibrinolysis. These factors act in concert to prevent thrombosis and to limit clotting to sites of vascular damage. However, if injured or exposed to proinflammatory factors, endothelial cells lose many of their antithrombotic properties. Here, we complete the discussion of hemostasis by focusing on the antithrombotic activities of normal endothelium; we return to the “dark side” of endothelial cells later when discussing thrombosis.

The antithrombotic properties of endothelium can be divided into activities directed at platelets, coagulation factors, and fibrinolysis.

- **Platelet inhibitory effects.** An obvious effect of intact endothelium is to serve as a barrier that shields platelets from subendothelial vWF and collagen. However, normal endothelium also releases several factors that inhibit platelet activation and aggregation. Among the most important are *prostacyclin* (PGI_2), *nitric oxide* (NO), and *adenosine diphosphatase*; the latter degrades ADP, already discussed as a potent activator of platelet aggregation. The major regulator of NO

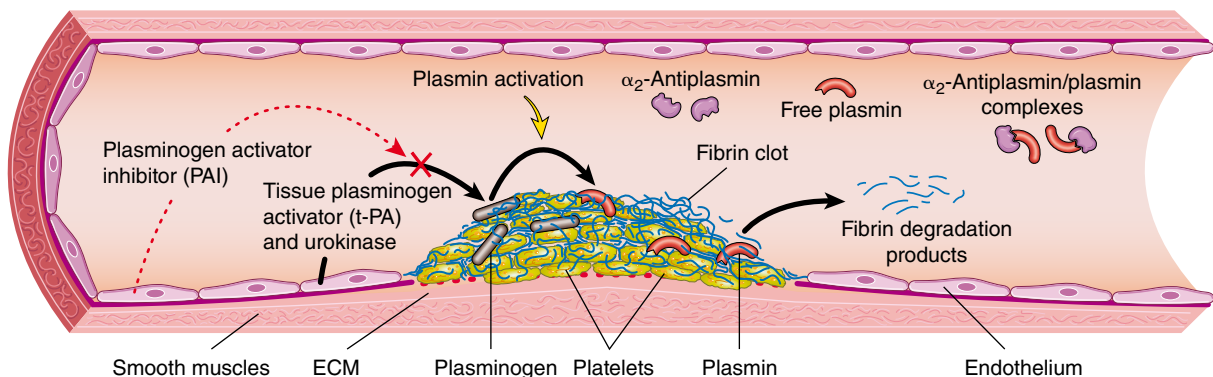


Fig. 4.9 Fibrinolytic system. Circulating plasminogen associates with fibrin clots and undergoes a change in conformation that makes it susceptible to activation by tissue plasminogen activator and urokinase. Active plasmin degrades fibrin and is in turn subject to inactivation by circulating α_2 -antiplasmin. Endothelium is also the source of plasminogen activators, negative regulators of plasmin activity. *ECM*, Extracellular matrix.

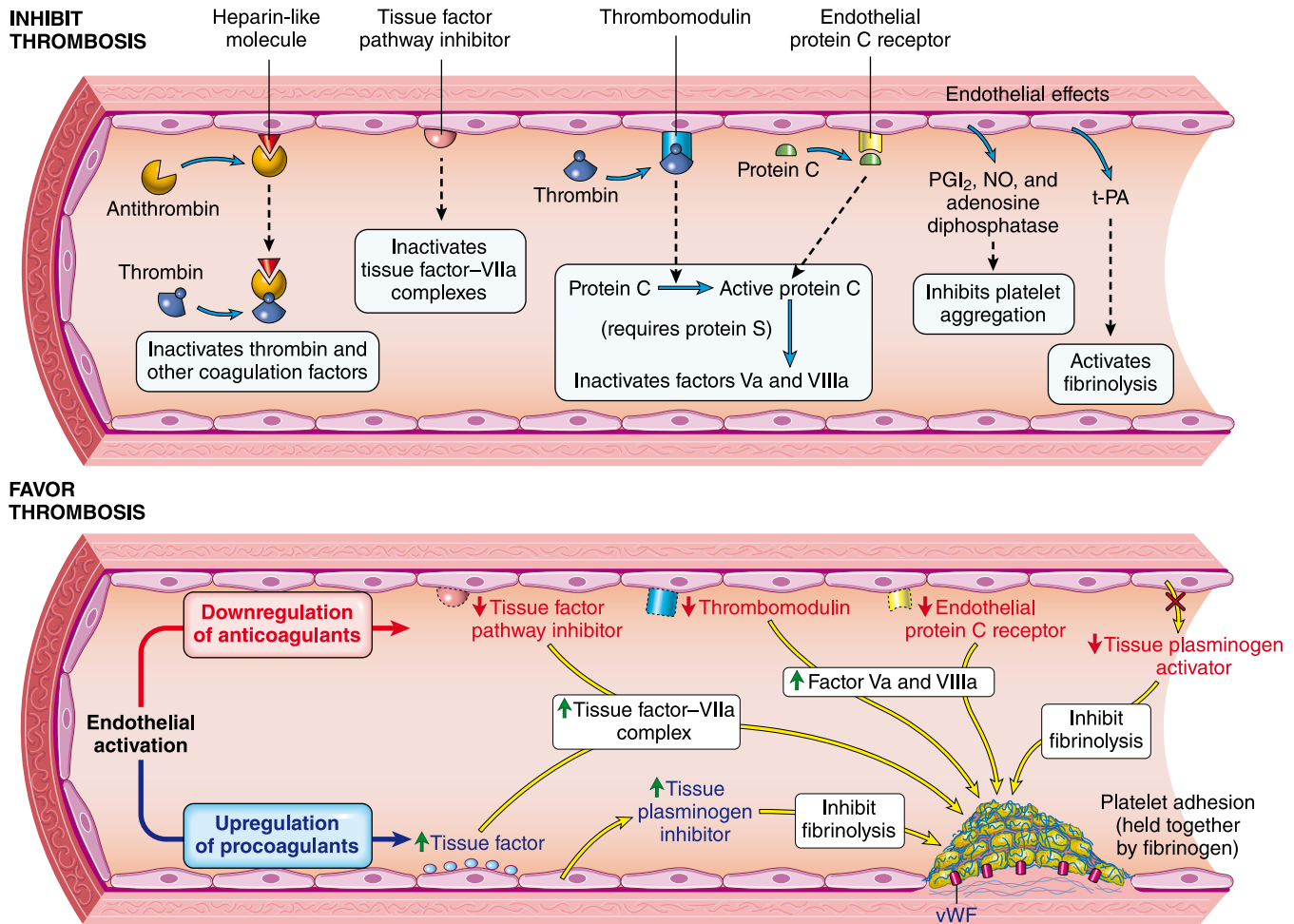


Fig. 4.10 Anticoagulant activities of normal endothelium (top) and procoagulant properties of injured or activated endothelium (bottom). NO, Nitric oxide; PGI₂, prostacyclin; t-PA, tissue plasminogen activator; vWF, von Willebrand factor. The thrombin receptor is one of the protease-activated receptors.

and PGI₂ production appears to be blood flow; precisely what senses flow is uncertain, though changes in cell shape and cytoskeleton are correlated. PGI₂ is produced by COX-1, which is expressed constitutively by “healthy” endothelium under normal flow conditions. NO is the product of endothelial nitric oxide synthase (eNOS).

- **Anticoagulant effects.** Normal endothelium shields coagulation factors from tissue factor in vessel walls and expresses multiple factors that actively oppose coagulation, most notably thrombomodulin, endothelial protein C receptor (EPCR), heparin-like molecules, and *tissue factor pathway inhibitor* (TFPI). Thrombomodulin and endothelial protein C receptor bind thrombin and protein C, respectively, in a complex on the endothelial cell surface. When bound in this complex, thrombin loses its ability to activate coagulation factors and platelets, and instead cleaves and activates protein C, a vitamin K-dependent protease that requires a cofactor, protein S. Activated protein C/protein S complex is a potent inhibitor of coagulation cofactors Va and VIIIa. Some studies suggest that protein S also exerts anticoagulation by inhibiting factor IX; serving as a cofactor for tissue factor pathway inhibitor; and interfering with factor VIIIa activity. Loss of these activities may contribute to the increased risk for abnormal clotting in those with protein S deficiency. *Heparin-like molecules* on the surface of endothelium bind and activate antithrombin III, which then inhibits thrombin and factors IXa, Xa, XIa,

and XIIa. The clinical utility of heparin and related drugs is based on their ability to stimulate antithrombin III activity. Tissue factor pathway inhibitor is synthesized in endothelial cells. Like protein C, it requires protein S as a cofactor and, as the name implies, binds and inhibits tissue factor/factor VIIa complexes.

- **Fibrinolytic effects.** Normal endothelial cells synthesize t-PA, as already discussed, a key component of the fibrinolytic pathway.

KEY CONCEPTS

Coagulation

- Coagulation occurs via the sequential enzymatic conversion of a cascade of circulating and locally synthesized proteins.
- Tissue factor elaborated at sites of injury is the most important initiator of the coagulation cascade in vivo.
- At the final stage of coagulation, thrombin converts fibrinogen into insoluble fibrin that contributes to formation of the definitive hemostatic plug.
- Coagulation normally is restricted to sites of vascular injury by:
 - Limiting enzymatic activation to phospholipid surfaces provided by activated platelets or endothelium
 - Circulating inhibitors of coagulation factors, such as antithrombin III, whose activity is augmented by heparin-like molecules expressed on endothelial cells

- Expression of additional factors on intact endothelial cells, such as thrombomodulin, tissue factor pathway inhibitor, endothelial protein C receptor, prostacyclin, NO, and adenosine diphosphatase, with anticoagulant activities
- Activation of fibrinolytic pathways (e.g., by association of t-PA with fibrin)

Hemorrhagic Disorders

Disorders associated with abnormal bleeding inevitably stem from primary or secondary defects in vessel walls, platelets, or coagulation factors, all of which must function properly to ensure hemostasis. The presentation of abnormal bleeding varies widely. At one end of the spectrum are massive bleeds associated with ruptures of large vessels such as the aorta or of the heart; these catastrophic events simply overwhelm hemostatic mechanisms and are often fatal. Diseases associated with sudden, massive hemorrhage include aortic dissection and aortic abdominal aneurysm (Chapter 11) and myocardial infarction (Chapter 12) complicated by rupture of the aorta or the heart. At the other end of the spectrum are subtle defects in clotting that only become evident under conditions of hemostatic stress, such as surgery, childbirth, dental procedures, menstruation, or trauma. Among the most common causes of mild bleeding tendencies are inherited defects in vWF (Chapter 14), aspirin consumption, and uremia (renal failure); the latter reduces platelet adhesion and aggregation. Between these extremes lie deficiencies of coagulation factors (hemophilias, Chapter 14), which are usually inherited and lead to severe bleeding disorders if untreated.

Additional specific examples of disorders associated with abnormal bleeding are discussed throughout the book. The following are general principles related to abnormal bleeding and its consequences.

- **Defects of primary hemostasis (platelet defects or von Willebrand disease)** often present with small bleeds in skin or mucosal membranes, which typically take the form of *petechiae*, minute 1- to 2-mm hemorrhages (Fig. 4.11A), or *purpura*, which are slightly larger (≥ 3 mm) than petechiae. It is believed that the capillaries of the mucosa and skin are particularly prone to rupture following minor trauma and that under normal circumstances platelets seal these defects virtually immediately. Mucosal bleeding associated with defects in primary hemostasis may also take the form of epistaxis (nosebleeds), gastrointestinal bleeding, or excessive menstruation (menorrhagia). A feared complication of very low platelet counts (*thrombocytopenia*) is intracerebral hemorrhage, which may be fatal.

- **Defects of secondary hemostasis (coagulation factor defects)** often present with bleeds into soft tissues (e.g., muscle) or joints. Bleeding into joints (*hemarthrosis*) following minor trauma is particularly characteristic of hemophilia (Chapter 14). It is unknown why severe defects in secondary hemostasis present with this peculiar pattern of bleeding; as with severe platelet defects, intracranial hemorrhage, sometimes fatal, may also occur.
- **Generalized defects involving small vessels** often present with “palpable purpura” and ecchymoses. *Ecchymoses* (sometimes simply called *bruises*) are hemorrhages of 1 to 2 cm in size. In both purpura and ecchymoses, the volume of extravasated blood may be large enough to create a palpable mass of blood known as a *hematoma*. Purpura and ecchymoses are particularly characteristic of systemic disorders that disrupt small blood vessels (e.g., vasculitis, Chapter 11) or that lead to blood vessel fragility (e.g., amyloidosis, Chapter 6; scurvy, Chapter 9).

The clinical significance of hemorrhage depends on the volume of the bleed, the rate at which it occurs, and its location. Rapid loss of up to 20% of the blood volume may have little impact in healthy adults; greater losses, however, can cause *hemorrhagic (hypovolemic) shock* (discussed later). Bleeding that is relatively trivial in the subcutaneous tissues can cause death if located in the brain (see Fig. 4.11B); because the skull is unyielding, intracranial hemorrhage may increase intracranial pressure to a level that compromises the blood supply or causes herniation of the brainstem (Chapter 28). Finally, chronic or recurrent external blood loss (e.g., peptic ulcer or menstrual bleeding) causes iron loss and can lead to iron deficiency anemia. In contrast, when red cells are retained (e.g., hemorrhage into body cavities or tissues), iron is recovered and recycled for use in the synthesis of hemoglobin.

Thrombosis

The primary abnormalities that lead to thrombosis are (1) endothelial injury, (2) stasis or turbulent blood flow, and (3) hypercoagulability of the blood (the so-called Virchow triad) (Fig. 4.12). Thrombosis is one of the scourges of modern man, as it underlies the most serious and common forms of cardiovascular disease. Here, the focus is on its causes and consequences; its role in cardiovascular disorders is discussed in detail in Chapters 11 and 12.

Endothelial Injury

Endothelial injury leading to platelet activation almost inevitably underlies thrombus formation in the heart and the arterial circulation,

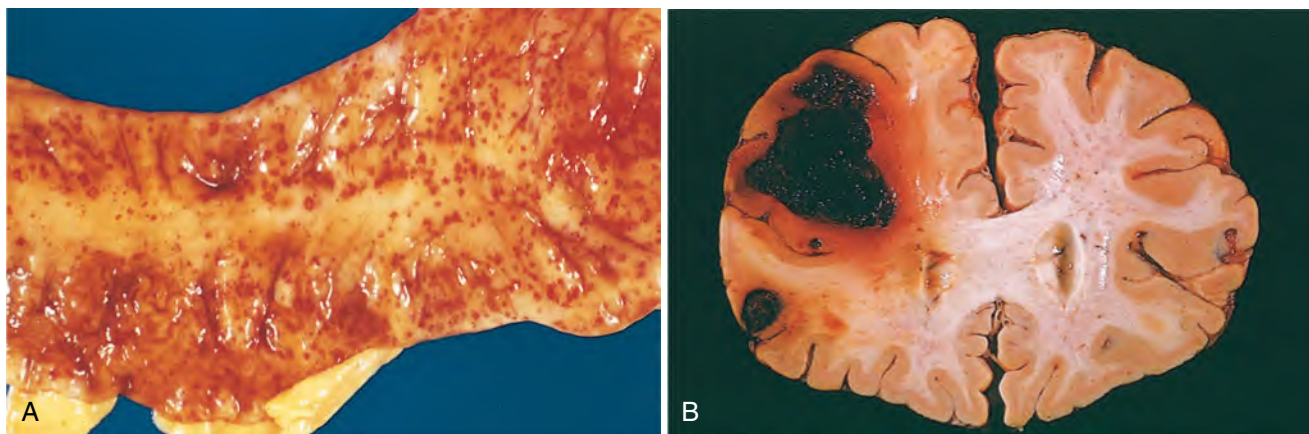


Fig. 4.11 (A) Punctate petechial hemorrhages of the colonic mucosa, a consequence of thrombocytopenia. (B) Fatal intracerebral bleed.

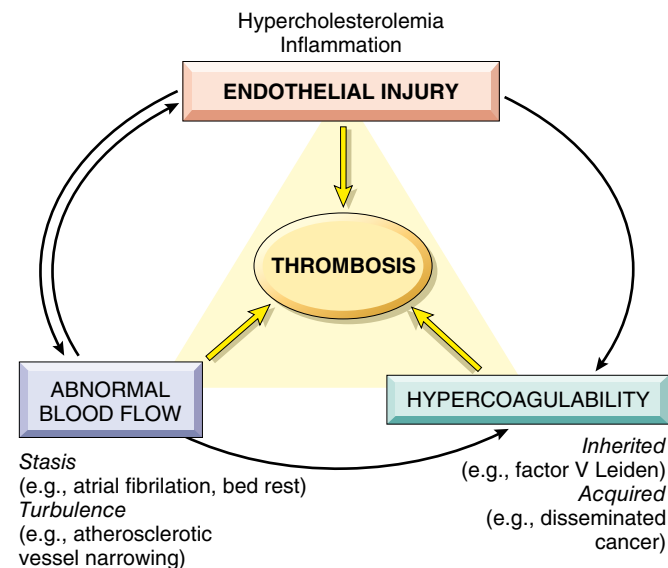


Fig. 4.12 Virchow triad in thrombosis. Endothelial integrity is the most important factor. Injury to endothelial cells can alter local blood flow and affect coagulability. Abnormal blood flow (stasis or turbulence), in turn, can cause endothelial injury. These factors may promote thrombosis independently or in combination.

where the high rates of blood flow impede clot formation. Notably, cardiac and arterial clots are typically rich in platelets, and it is believed that platelet adherence and activation is a necessary prerequisite for thrombus formation under high shear stress, such as exists in arteries. This insight provides part of the reasoning behind the use of aspirin and other platelet inhibitors in coronary artery disease and acute myocardial infarction.

Obviously, severe endothelial injury may trigger thrombosis by exposing vWF and tissue factor. However, inflammation and other noxious stimuli also promote thrombosis by shifting the pattern of gene expression in endothelium to one that is “prothrombotic.” This change is sometimes referred to as *endothelial activation* or *dysfunction* and can be produced by diverse exposures, including physical injury, infectious agents, abnormal blood flow, inflammatory mediators, metabolic abnormalities such as hypercholesterolemia or homocystinemia, and toxins absorbed from cigarette smoke. Endothelial activation is believed to have an important role in triggering arterial thrombotic events.

The role of endothelial cell activation and dysfunction in arterial thrombosis is discussed in detail in Chapters 11 and 12. Here it suffices to mention several of the major prothrombotic alterations (see Fig 4.10, bottom):

- **Procoagulant changes.** Endothelial cells activated by inflammatory cytokines downregulate the expression of *thrombomodulin*, already described as a key modulator of thrombin activity, enhancing the procoagulant and proinflammatory actions of thrombin. In addition, inflamed endothelium downregulates the expression of other anticoagulants such as protein C and tissue factor protein inhibitor, changes that further promote a procoagulant state.
- **Antifibrinolytic effects.** Activated endothelial cells secrete *plasminogen activator inhibitors* (PAIs), which limit fibrinolysis, and downregulate the expression of t-PA, alterations that favor the development of thrombi.

Alterations in Normal Blood Flow

Turbulence contributes to arterial and cardiac thrombosis by causing endothelial injury or dysfunction and by forming countercurrents

that contribute to local pockets of stasis; by contrast, stasis is a major contributor to the development of venous thrombi. Normal blood flow is laminar such that the platelets (and other blood cellular elements) flow centrally in the vessel lumen, separated from the endothelium by a slower moving layer of plasma. Turbulence and stasis therefore:

- *Promote endothelial activation*, enhancing procoagulant activity and leukocyte adhesion, in part through flow-induced changes in the expression of adhesion molecules and proinflammatory factors
- *Disrupt laminar flow* and bring platelets into contact with the endothelium
- *Prevent washout and dilution of activated clotting factors* by fresh flowing blood and the inflow of clotting factor inhibitors

Altered blood flow contributes to thrombosis in several clinical settings. Ulcerated atherosclerotic plaques not only expose subendothelial vWF and tissue factor but also cause turbulence. Aortic and arterial dilations called *aneurysms* result in local stasis and are therefore fertile sites for thrombosis (Chapter 11). Acute myocardial infarction results in areas of noncontractile myocardium and sometimes in cardiac aneurysm; both are associated with stasis and flow abnormalities that promote the formation of cardiac mural thrombi (Chapter 12). Rheumatic mitral valve stenosis results in left atrial dilation; in conjunction with atrial fibrillation, a dilated atrium is a site of profound stasis and a prime location for thrombosis (Chapter 12). Hyperviscosity (such as is seen with polycythemia vera; Chapter 13) increases resistance to flow and causes small vessel stasis, and the deformed red cells in *sickle cell anemia* (Chapter 14) impede blood flow through small vessels, with the resulting stasis also predisposing to thrombosis.

Hypercoagulability

Hypercoagulability refers to an abnormally high tendency of the blood to clot and is typically caused by alterations in coagulation factors. Hypercoagulability has a particularly important role in venous thrombosis and can be divided into primary (genetic) and secondary (acquired) disorders (Table 4.2). Of the inherited causes of hypercoagulability, point mutations in the factor V gene and prothrombin gene are the most common:

- **Factor V Leiden.** Approximately 2% to 15% of people with European ancestry carry a single-nucleotide mutation in factor V that is called *factor V Leiden*, after the city in the Netherlands where it was discovered. Among individuals with recurrent DVT, the frequency of this mutation is considerably higher, approaching 60%. This mutation renders factor V resistant to cleavage and inactivation by protein C. As a result, an important antithrombotic counterregulatory pathway is lost (see Fig. 4.10). The inheritance pattern for factor V Leiden is autosomal dominant. Heterozygotes have a 4- to 8-fold increased relative risk of venous thrombosis, and homozygotes have an approximately 12-fold increased risk.
- **Prothrombin gene mutation.** A single nucleotide change (G20210A) in the 3'-untranslated region of the prothrombin gene is another common mutation (1% to 2% of the population) associated with hypercoagulability. It leads to elevated prothrombin levels and a two- to four-fold increased risk of venous thrombosis in heterozygotes and seven-fold increased risk in homozygotes.
- **Other inherited causes.** Much less common causes of primary hypercoagulability include deficiencies of anticoagulants such as antithrombin III, protein C, or protein S; affected individuals typically present with venous thrombosis and recurrent thromboembolism beginning in adolescence or early adulthood.
- **Homocystinemia.** Elevated levels of homocysteine may be inherited or acquired. Marked elevations of homocysteine may be caused by an inherited deficiency of *cystathione β -synthetase*.

TABLE 4.2 Hypercoagulable States

Primary (Genetic)
Common
- Factor V mutation: factor V Leiden (Arg to Gln substitution in amino acid residue 506 leading to resistance to activated protein C) - Prothrombin mutation (G20210A noncoding sequence variant leading to increased prothrombin levels) - Increased levels of factors VIII, IX, XI, or fibrinogen (genetics unknown)
Rare
- Antithrombin III deficiency - Protein C deficiency - Protein S deficiency
Very Rare
- Fibrinolysis defects - Homozygous homocystinuria (deficiency of cystathione β-synthetase)
Secondary (Acquired)
Strong Risk Factors for Thrombosis
- Prolonged bed rest or immobilization - Myocardial infarction - Atrial fibrillation - Tissue injury (surgery, fracture, burn) - Cancer - Prosthetic cardiac valves - Disseminated intravascular coagulation - Heparin-induced thrombocytopenia - Antiphospholipid antibody syndrome
Other Risk Factors for Thrombosis
- Cardiomyopathy - Nephrotic syndrome - Hyperestrogenic states (pregnancy and postpartum) - Oral contraceptive use - Sickle cell anemia - Smoking

Acquired causes include deficiency of vitamin B₆, B₁₂, and folic acid. Prothrombotic effects of homocysteine have been reported in some but not all studies.

The most common thrombophilic genotypes found in various populations (heterozygosity for factor V Leiden and heterozygosity for the prothrombin G20210A variant) impart only a moderately increased risk of thrombosis; most individuals with these genotypes, when otherwise healthy, are free from thrombotic complications. However, factor V and prothrombin mutations are frequent enough that homozygosity and compound heterozygosity are not rare, and such genotypes are associated with greater risk. Moreover, individuals with such mutations have a significantly increased frequency of venous thrombosis in the setting of other acquired risk factors (e.g., pregnancy or prolonged bed rest). Thus, factor V Leiden heterozygosity may trigger DVT when combined with enforced inactivity, such as during prolonged airplane travel. Consequently, **inherited causes of hypercoagulability must be considered in patients younger than 50 years who present with thrombosis—even when acquired risk factors are present.**

Unlike hereditary disorders, the pathogenesis of *acquired thrombophilia* is frequently multifactorial (see Table 4.2). In some cases (e.g., cardiac failure or trauma), stasis or vascular injury may be most important. Hypercoagulability due to oral contraceptive use or the hyperestrogenic state of pregnancy is probably caused by increased hepatic

synthesis of coagulation factors and reduced anticoagulant synthesis. In disseminated cancers, release of various procoagulants from tumors predisposes to thrombosis. The hypercoagulability seen with advancing age may be due to reduced endothelial PGI₂ production. Smoking and obesity promote hypercoagulability by unknown mechanisms.

Among the acquired thrombophilic states, heparin-induced thrombocytopenia and antiphospholipid antibody syndrome are particularly important clinical problems that deserve special mention.

Heparin-Induced Thrombocytopenia Syndrome. *Heparin-induced thrombocytopenia (HIT) syndrome is a serious, potentially life-threatening disorder that occurs following the administration of unfractionated heparin.* HIT results from the formation of antibodies that recognize complexes of heparin and PF4. Platelets store PF4 in alpha granules, from which it is released gradually during their 8- to 10-day lifespan or acutely with platelet activation and degranulation. Released PF4 binds reversibly to glycosaminoglycan molecules on the surface of endothelial cells. When heparin is injected, it displaces PF4 from endothelium, and forms PF4/heparin complexes that persist for some time in the circulation. Importantly, binding of heparin also causes a conformational change in PF4 that creates a neoantigen against which IgG antibodies are formed. By mechanisms that remain to be defined, it is thought that PF4-IgG complexes attach to Fc receptors on platelet surfaces, leading to their activation. In turn, activated platelets express procoagulant molecules, both secreted and platelet-bound, that promote thrombosis. Binding of anti-PF4 antibodies to platelets also results in their removal by macrophages (contributing to the thrombocytopenia in the syndrome name). Thrombocytopenia is the most common manifestation, but thrombosis is far more serious. It occurs in approximately 50% of cases and affects both veins and arteries. Necrosis of the skin, gangrene of the limbs, stroke, and myocardial infarction are some of the sequelae. Diagnosis requires the demonstration of anti-PF4-heparin antibodies. Low-molecular-weight heparin preparations induce HIT less frequently, and other classes of anticoagulants such as direct inhibitors of factor X or thrombin avoid the risk altogether.

Vaccine-induced thrombotic thrombocytopenia (VITT) is an extremely rare complication that can occur in those who received certain adenoviral-vector-based vaccines for coronavirus disease 2019 (COVID-19). As with HIT, VITT is characterized by antibodies to PF4. Instead of heparin, however, PF4 is rendered immunogenic by binding to some adenoviral proteins. Thrombosis may involve cerebral venous sinuses (with hemorrhage), deep veins of the leg, pulmonary arteries, and splanchnic vessels. VITT can be a life-threatening disorder requiring prompt diagnosis and treatment.

Antiphospholipid Antibody Syndrome. Antiphospholipid syndrome (APS) is an autoimmune disorder characterized by:

- Presence of one or more antiphospholipid (aPL) autoantibodies.
- Venous or arterial thromboses, or pregnancy complications such as recurrent miscarriages, unexplained fetal death, and premature birth.

Antiphospholipid antibody syndrome has primary and secondary forms. Individuals with a well-defined autoimmune disease, such as systemic lupus erythematosus (SLE) (Chapter 6), are designated as having *secondary antiphospholipid syndrome*. Because of the association with lupus and the ability of these antibodies to inhibit clotting in vitro, they are referred to as the “lupus anticoagulant.”

In *primary antiphospholipid syndrome*, patients exhibit only the manifestations of a hypercoagulable state and lack evidence of other

well-defined autoimmune disorders. Approximately 50% of patients with APS have the primary form, and the rest occur in association with an autoimmune disease, most commonly systemic lupus erythematosus. Our focus here is on primary APS.

The clinical manifestations of APS are varied; they include recurrent thromboses, repeated miscarriages, cardiac valve vegetations, and thrombocytopenia. Depending on the vascular bed involved, the clinical presentations can include pulmonary embolism (following lower extremity venous thrombosis), pulmonary hypertension (from recurrent subclinical pulmonary emboli), valvular heart disease, stroke, bowel infarction, or renovascular hypertension.

The pathogenesis of antiphospholipid syndrome is complex and not fully understood. The aPL antibodies are directed against anionic membrane phospholipids or proteins associated with phospholipids. Proteins that are recognized by these antibodies include cardiolipin and β_2 -glycoprotein I. The latter is found in plasma, but it has strong avidity for phospholipids expressed on the surfaces of endothelial cells, monocytes, platelets, thrombin, and trophoblasts. Anti- β_2 -glycoprotein antibodies are suspected to have a major role in APS by activating endothelial cells, neutrophils, monocytes, and platelets. Their pathogenicity is supported by the observation that transfer of these antibodies into rodents can induce thrombosis. Patients with APS also show evidence of complement activation and inhibition of fibrinolytic processes, both of which favor the prothrombotic state. As mentioned earlier, pregnancy morbidity is a defining manifestation of APS. However, unlike most other clinical features, early fetal loss does not appear to be caused by thrombosis, but rather seems to stem from antibody-mediated interference with the growth and differentiation of trophoblasts, leading to a failure of placentation. By contrast, late complications of pregnancy such as preeclampsia and preterm birth, are caused both by trophoblast dysfunction and direct inflammatory effects mediated by antiphospholipid antibodies and complement.

Although antiphospholipid antibodies are clearly associated with thrombotic diatheses, they have also been identified in 5% to 15% of apparently normal individuals, implying that they are not sufficient to cause the full-blown syndrome. It is postulated that a “second hit” is required that may be provided by infection, smoking, or pregnancy, among others. Although these antibodies induce a hypercoagulable state in vivo, they interfere with phospholipids and thus inhibit coagulation in vitro, thereby prolonging the PTT, hence the name lupus anticoagulant. The antibodies also frequently result in a false-positive serologic test for syphilis because the antigen in the standard assay contains cardiolipin, which cross reacts with phospholipids of *Treponema pallidum*. Diagnosis of APS is based on clinical features and demonstration of antiphospholipid antibodies in the serum. Therapy involves various forms of anticoagulation.

MORPHOLOGY

Thrombi can develop anywhere in the cardiovascular system and vary in size and shape depending on the involved site and the underlying cause. Arterial or cardiac thrombi usually begin at sites of turbulence or endothelial injury, whereas venous thrombi characteristically occur at sites of stasis. Thrombi are focally attached to the underlying vascular surface, particularly at the point of initiation. From here, arterial thrombi tend to grow retrograde, and venous thrombi extend in the direction of blood flow; thus, both propagate toward the heart. The propagating portion of a thrombus is often poorly attached and therefore prone to fragmentation and embolization.

Thrombi often have grossly and microscopically apparent laminations called **lines of Zahn**, which are pale platelet and fibrin deposits alternating with darker red cell-rich layers. Such laminations signify that a thrombus has

formed in flowing blood; their presence can therefore distinguish antemortem clots from the bland nonlaminated clots that occur postmortem (see later).

Thrombi occurring in heart chambers or in the aortic lumen are designated **mural thrombi**. Abnormal myocardial contraction (arrhythmias, dilated cardiomyopathy, or myocardial infarction) or endomyocardial injury (myocarditis or catheter trauma) promotes cardiac mural thrombi (Fig. 4.13A) and ulcerated atherosclerotic plaque and aneurysmal dilation underlie aortic thrombi (Fig. 4.13B).

Arterial thrombi are frequently occlusive; the most common sites in decreasing order of frequency are the coronary, cerebral, and femoral arteries. They typically consist of a friable meshwork of platelets, fibrin, red cells, and degenerating leukocytes. Although these are usually superimposed on a ruptured atherosclerotic plaque, other vascular injuries (vasculitis, trauma) may be the underlying cause.

Venous thrombosis (phlebothrombosis) is almost invariably occlusive, with the thrombus forming a long luminal cast. Because these thrombi form in the sluggish venous circulation, they tend to contain more enmeshed red cells (and relatively few platelets) and are therefore known as **red thrombi** or **stasis thrombi**. Venous thrombi are firm, are focally attached to the vessel wall, and contain lines of Zahn, features that help distinguish them from postmortem clots (see later). The veins of the lower extremities are most commonly involved (90% of cases); however, upper extremities, periprostatic plexus, or the ovarian and peritestic veins can also develop venous thrombi. Under special circumstances, they can also occur in the dural sinuses, portal vein, or hepatic vein.

Postmortem clots can sometimes be mistaken for antemortem venous thrombi. However, clots that form after death are gelatinous and have a dark-red dependent portion where red cells have settled by gravity and a yellow “chicken fat” upper portion and are usually not attached to the underlying vessel wall.

Thrombi on heart valves are called **vegetations**, which may be infected or sterile. Blood-borne bacteria or fungi can adhere to previously damaged valves (e.g., due to rheumatic heart disease) or may cause valve damage directly; in either case, endothelial injury and disturbed blood flow can induce the formation of large thrombotic masses (**infective endocarditis**; Chapter 12). Sterile vegetations can also develop on noninfected valves in persons with hypercoagulable states, so-called **nonbacterial thrombotic endocarditis** (Chapter 12). Less commonly, sterile verrucous endocarditis (**Libman-Sacks endocarditis**) can occur in the setting of systemic lupus erythematosus (Chapter 6).

Fate of the Thrombus

If a patient survives the initial thrombosis, in the ensuing days to weeks thrombi undergo some combination of four events:

- **Propagation.** Thrombi accumulate additional platelets and fibrin and enlarge (discussed earlier).
- **Embolization.** Thrombi dislodge and travel to other sites in the vasculature (discussed later).
- **Dissolution.** Dissolution is the result of fibrinolysis, which can lead to the rapid shrinkage and total disappearance of recent thrombi. In contrast, extensive fibrin deposition and cross-linking in older thrombi render them more resistant to lysis. This distinction explains in part why therapeutic administration of fibrinolytic agents such as t-PA (e.g., in the setting of acute coronary thrombosis) is generally effective only when given during the first few hours of a thrombotic event.
- **Organization and recanalization.** Older thrombi become organized by the ingrowth of endothelial cells, smooth muscle cells, and fibroblasts (Fig. 4.14). Capillary channels that reestablish the continuity of the original lumen, albeit to a variable degree, eventually form.

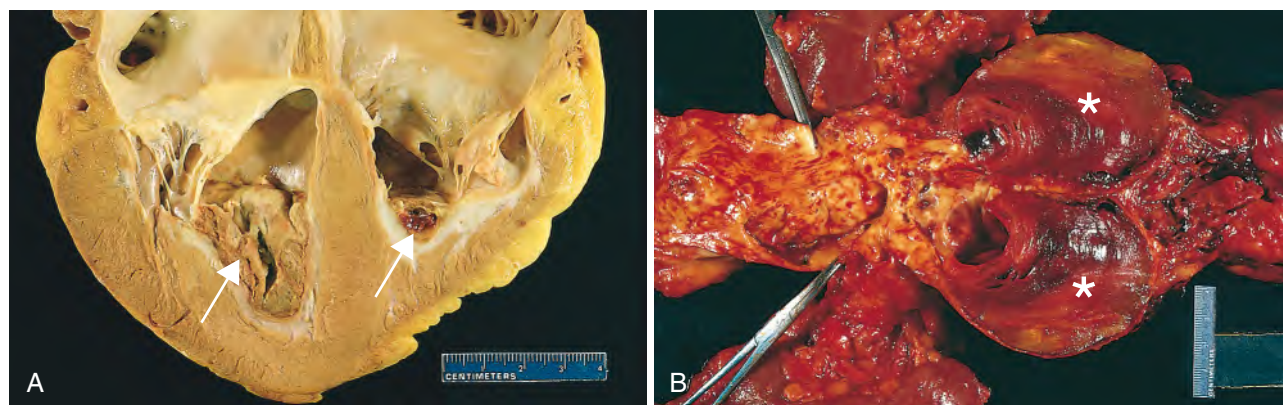


Fig. 4.13 Mural thrombi. (A) Thrombus in the left and right ventricular apices (*arrows*), overlying a white fibrous scar. (B) Laminated thrombus in a dilated abdominal aortic aneurysm (*asterisks*). Numerous friable mural thrombi are also superimposed on advanced atherosclerotic lesions of the more proximal aorta (*left side of image*).

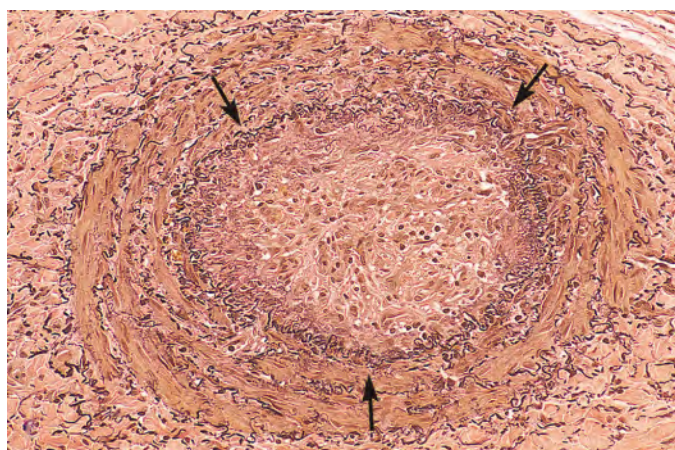


Fig. 4.14 Low-power view of a thrombosed artery stained for elastic tissue. The original lumen is delineated by the internal elastic lamina (*arrows*) and is totally filled with organized thrombus.

Continued recanalization may convert a thrombus into a smaller mass of connective tissue that becomes incorporated into the vessel wall. Eventually, with remodeling and contraction of the mesenchymal elements, only a fibrous lump may remain to mark the original thrombus (Chapter 11).

Occasionally the centers of thrombi undergo enzymatic digestion, presumably as a result of the release of lysosomal enzymes from trapped leukocytes and platelets. In the setting of bacteremia, such thrombi may become infected, producing an inflammatory mass that erodes and weakens the vessel wall. If unchecked, this may result in a mycotic aneurysm (Chapter 11).

Clinical Features

Thrombi come to clinical attention when they obstruct arteries or veins or give rise to emboli. The clinical presentation depends on the involved site. Venous thrombi can cause painful congestion and edema distal to an obstruction but are mainly of concern due to their tendency to embolize to the lungs (see later). Conversely, although arterial thrombi

can also embolize and cause downstream infarctions, the chief clinical problem is more often related to occlusion of a critical vessel (e.g., a coronary or cerebral artery), which can have serious or fatal consequences.

Venous Thrombosis (Phlebothrombosis). Most venous thrombi occur in the superficial or deep veins of the leg. Superficial venous thrombi typically occur in the saphenous veins in the setting of varicosities. Such thrombi can cause local congestion, swelling, pain, and tenderness, but rarely embolize. Nevertheless, the associated edema and impaired venous drainage predispose the overlying skin to the development of infection and ulcers (*varicose ulcers*). Deep vein thrombosis (DVT) involving one of the large leg veins—at or above the knee (e.g., the popliteal, femoral, and iliac veins)—is considered serious because such thrombi more often embolize to the lungs and give rise to pulmonary infarction (see later and Chapter 15). Although DVTs may cause local pain and edema due to venous obstruction, these symptoms are often absent due to the opening of venous collateral channels. Consequently, DVTs are asymptomatic in approximately 50% of affected individuals and are recognized only in retrospect after embolization.

Lower extremity DVTs are often associated with hypercoagulable states, as described earlier (see Table 4.2). Common predisposing factors include bed rest and immobilization (because they reduce the milking action of the leg muscles, resulting in stasis), and congestive heart failure (also a cause of impaired venous return). Trauma, surgery, and burns not only immobilize a person but are also associated with vascular insults, procoagulant release from injured tissues, increased hepatic synthesis of coagulation factors, and decreased t-PA production. Many elements contribute to the heightened thrombotic risk during pregnancy, including decreased venous return from leg veins and systemic hypercoagulability associated with the hormonal changes of late pregnancy and the postpartum period. Tumor-associated inflammation and release of coagulation factors (e.g., tissue factor) and procoagulants (e.g., mucin) from tumor cells all contribute to the increased risk of thromboembolism in disseminated cancers, so-called *migratory thrombophlebitis* or *Trousseau syndrome*. Regardless of the specific clinical setting, advanced age also increases the risk of DVT.

Arterial and Cardiac Thrombosis. Atherosclerosis is a major cause of arterial thromboses because it is associated with loss of endothelial integrity and with abnormal blood flow (see Fig. 4.13B). Myocardial infarction can predispose to cardiac mural thrombi by causing dyskinetic myocardial contraction and endocardial injury (see Fig. 4.13A),

and rheumatic heart disease may engender atrial mural thrombi by causing atrial dilation and fibrillation. Both cardiac and aortic mural thrombi are prone to embolization. Although any tissue can be affected, the brain, kidneys, and spleen are particularly likely targets because of their rich blood supply.

KEY CONCEPTS

Thrombosis

- Thrombus development usually is related to one or more components of the Virchow triad:
 - Endothelial injury (e.g., by toxins, hypertension, inflammation, or metabolic products) associated with endothelial activation and changes in endothelial gene expression that favor coagulation
 - Abnormal blood flow—stasis or turbulence (e.g., due to aneurysms, atherosclerotic plaque)
 - Hypercoagulability, either primary (e.g., factor V Leiden, increased prothrombin synthesis, antithrombin deficiency) or secondary (e.g., bed rest, tissue damage, malignancy, or development of antiphospholipid antibodies [antiphospholipid antibody syndrome]) or antibodies against PF4/heparin complexes [heparin-induced thrombocytopenia])
- Thrombi may propagate, resolve, organize, or embolize.
- Thrombosis causes tissue injury by local vascular occlusion or by distal embolization.

Disseminated Intravascular Coagulation

Disseminated intravascular coagulation (DIC) is widespread thrombosis within the microcirculation that may be of sudden or insidious onset. It is not a specific disease but rather a complication of many conditions associated with systemic activation of thrombin. Disorders ranging from obstetric complications to advanced malignancy can be complicated by DIC, which leads to widespread formation of thrombi in the microcirculation. These microvascular thrombi can cause diffuse circulatory insufficiency and organ dysfunction, particularly of the brain, lungs, heart, and kidneys. To complicate matters, the runaway thrombosis “uses up” platelets and coagulation factors (hence the synonym *consumptive coagulopathy*) and often activates fibrinolytic mechanisms. Thus, symptoms initially related to thrombosis can evolve into a bleeding catastrophe, such as hemorrhagic stroke or hypovolemic shock. DIC is discussed in greater detail along with other bleeding disorders in Chapter 14.

EMBOLISM

An embolus is a detached intravascular solid, liquid, or gaseous mass that is carried by the blood from its point of origin to a distant site, where it often obstructs blood flow and causes tissue dysfunction or infarction. Most emboli are dislodged thrombi, hence the term *thromboembolism*. Other rare emboli are composed of fat droplets, air or nitrogen bubbles, atherosclerotic debris (*cholesterol emboli*), tumor fragments, bone marrow, or even foreign bodies. Emboli travel through the blood until they encounter vessels too small to permit further passage, causing partial or complete vascular occlusion. Depending on where they originate, emboli can lodge anywhere in the vascular tree; as discussed later, the clinical consequences vary widely depending on the size and the position of the lodged embolus, as well as the vascular bed that is impacted.

Pulmonary Embolism

Pulmonary emboli originate from DVT and are a common, serious form of thromboembolic disease. Pulmonary embolism (PE) has an estimated incidence of 60 to 120 cases per 100,000 people in the United States. It is somewhat more common in males than in females. PE causes 60,000 to 100,000 deaths per year in the United States. An estimated 20% of individuals with PE die before or shortly after a diagnosis is made. In 70% to 80% of cases, PE originates from leg DVT. Hence the risk factors for PE are the same as for DVT (see Table 4.2).

Fragmented thrombi from DVT are carried through progressively larger veins and the right side of the heart before slamming into the pulmonary arterial vasculature. Depending on the size of the embolus, it can occlude the main pulmonary artery, straddle the pulmonary artery bifurcation (*saddle embolus*), or pass out into the smaller, branching arteries (Fig. 4.15). Frequently there are multiple emboli, occurring either sequentially or simultaneously as a shower of smaller emboli from a single large mass; in general, **the patient who has had one PE is at high risk for more.** Rarely, a venous embolus passes through an interatrial or interventricular defect and gains access to the systemic arterial circulation (*paradoxical embolism*). A more complete discussion of PEs is presented in Chapter 15; the following is an overview of the major functional consequences of pulmonary emboli.

- *Most pulmonary emboli (60% to 80%) are clinically silent* because they are small. With time they become organized and are incorporated into the vascular wall; in some cases, organization of the thromboembolus leaves behind a delicate, bridging fibrous web.
- *Sudden death, acute right-heart failure (cor pulmonale), or cardiovascular collapse occurs* when emboli obstruct 60% or more of the pulmonary circulation.
- *Embolic obstruction of medium-sized arteries with subsequent vascular rupture can result in pulmonary hemorrhage* but usually does not cause pulmonary infarction. This is because the lung is supplied by both the pulmonary arteries and the bronchial arteries, and the intact bronchial circulation is usually sufficient to perfuse the affected area. Understandably, if the bronchial arterial flow is compromised (e.g., by left-sided cardiac failure), infarction may occur.
- *Embolic obstruction of small end-arteriolar pulmonary branches often does produce hemorrhage or infarction.*
- *Multiple emboli over time may cause pulmonary hypertension and right ventricular failure.*

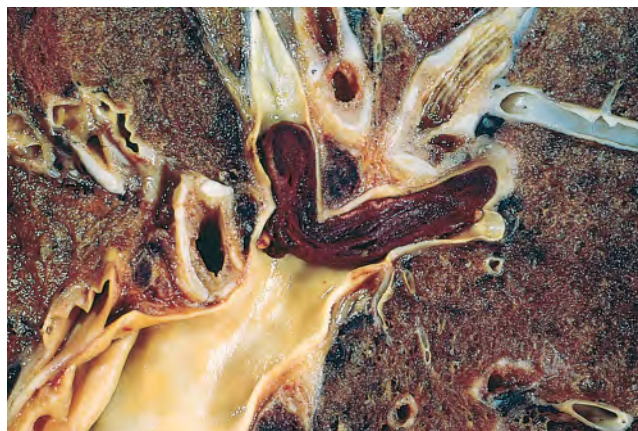


Fig. 4.15 Embolus from a lower extremity deep venous thrombosis, lodged at a pulmonary artery branchpoint.

Systemic Thromboembolism

Most systemic emboli (80%) arise from intracardiac mural thrombi, two-thirds of which are associated with left ventricular wall infarcts and another one-fourth with left atrial dilation and fibrillation. The remainder originate from aortic aneurysms, atherosclerotic plaques, valvular vegetations, or venous thrombi (paradoxical emboli); 10% to 15% are of unknown origin. In contrast to venous emboli, the vast majority of which lodge in the lung, arterial emboli can travel to a wide variety of sites; the point of arrest depends on the source and the relative amount of blood flow that downstream tissues receive. Most come to rest in the lower extremities (75%) or the brain (10%), but other tissues, including the intestines, kidneys, spleen, and upper extremities, may be involved. The consequences of systemic emboli depend on the vulnerability of the affected tissues to ischemia, the caliber of the occluded vessel, and whether a collateral blood supply exists; in general, however, the outcome is tissue infarction.

Fat Embolism

Fat embolism refers to the presence of microscopic fat globules—sometimes with associated hematopoietic bone marrow—in the vasculature, typically after fractures of long bones or, rarely, in the setting of soft tissue trauma and burns. It is common, occurring in some 90% of individuals with severe skeletal injuries (Fig. 4.16). Presumably these injuries rupture vascular sinusoids or small venules in the marrow, allowing marrow and adipose tissue to herniate into the vascular space and travel to the lung. Fat embolism syndrome is the term applied to the minority of patients who become symptomatic.

Fat embolism syndrome is characterized by pulmonary insufficiency, neurologic symptoms, anemia, and thrombocytopenia, and is fatal in 5% to 15% of cases. Typically, 1 to 3 days after injury, there is a sudden onset of tachypnea, dyspnea, tachycardia, irritability, and restlessness, which ultimately can progress to delirium or coma. Thrombocytopenia is attributed to platelet adhesion to fat globules and subsequent aggregation or splenic sequestration; anemia can result from red cell aggregation and/or hemolysis. A diffuse petechial rash (seen in 20% to 50% of cases) is related to rapid onset of thrombocytopenia and can be a useful diagnostic clue.

The pathogenesis of fat embolism syndrome involves both mechanical obstruction and biochemical injury. Fat microemboli and

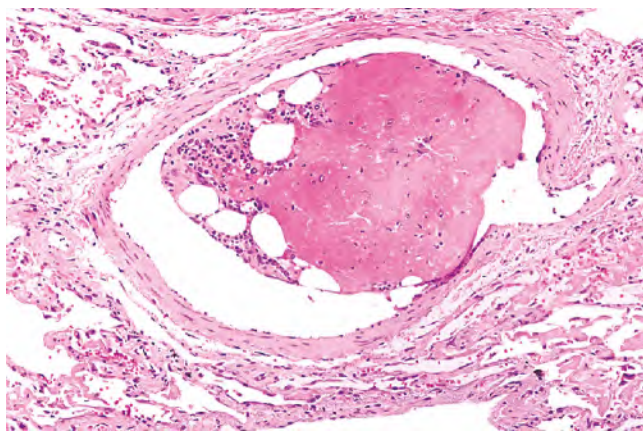


Fig. 4.16 Bone marrow embolus in the pulmonary circulation. The cellular elements on the left side of the embolus are hematopoietic cells, and the cleared vacuoles represent marrow fat. The relatively uniform red area on the right of the embolus is an early organizing thrombus.

associated red cell and platelet aggregates can occlude the pulmonary and cerebral microvasculature. Release of free fatty acids from the fat globules exacerbates the situation by causing local toxic injury to endothelium, and platelet activation and granulocyte recruitment (with free radical, protease, and eicosanoid release) complete the vascular assault. Because lipids are dissolved out of tissue preparations by solvents routinely used in paraffin embedding, the microscopic demonstration of fat microglobules, in the absence of accompanying marrow elements, typically requires specialized techniques, including frozen sections and stains for fat.

Air Embolism

Gas bubbles within the circulation can coalesce to form frothy masses that obstruct vascular flow and cause distal ischemic injury.

Air embolism occurs when there is communication between the vasculature and outside air and a negative pressure gradient that “sucks” in the air. For example, air may be introduced into the cerebral circulation by neurosurgery in the “sitting position,” creating a gravitational gradient. Air may also be introduced during endovascular and interventional procedures as well as during mechanical ventilation. A large volume of air, generally more than 100 mL, is necessary to produce a clinical effect in the pulmonary circulation; unless care is taken, this volume of air can be inadvertently introduced during obstetric or laparoscopic procedures, or because of chest wall injury. Introduction of 300 to 500 mL of air over several seconds may be fatal. Entry of air in the pulmonary vasculature not only blocks perfusion of downstream region but also induces an intense inflammatory response with release of cytokines that may injure the alveoli. Bubbles in the central nervous system can cause memory loss, ataxia, visual disturbances, and even the sudden onset of coma.

A particular form of gas embolism, called *decompression sickness*, occurs when individuals experience sudden decreases in atmospheric pressure. Scuba and deep-sea divers, underwater construction workers, and persons in unpressurized aircraft who undergo rapid ascent are all at risk. Care must also be taken to avoid rapid decompression during space travel. Upon breathing air at high pressure (e.g., during a deep-sea dive), increased amounts of gas (particularly nitrogen) are dissolved in the blood and tissues. If the diver then ascends (depressurizes) too rapidly, the nitrogen comes out of solution in the tissues and the blood.

The rapid formation of gas bubbles within skeletal muscles and supporting tissues in and about joints is responsible for the painful condition called *the bends* (so named in the 1880s because it was noted that those afflicted characteristically arched their backs in a manner reminiscent of a then-popular women’s fashion pose called the *Grecian bend*). Even small intravascular bubbles may have physical effects, caused by inflammatory and thrombogenic host responses. In the lungs, gas bubbles in the vasculature cause edema, hemorrhage, and focal atelectasis or emphysema, leading to a form of respiratory distress called *the chokes*. A more chronic form of decompression sickness is called *caisson disease* (named for the pressurized vessels used in bridge construction; workers in these vessels suffered both acute and chronic forms of decompression sickness). In caisson disease, persistence of gas emboli in the skeletal system leads to multiple foci of ischemic necrosis; the more common sites are the femoral heads, tibia, and humeri.

Individuals affected by acute decompression sickness are treated by being placed in a chamber under sufficiently high pressure to force the gas bubbles back into solution. Subsequent slow decompression permits gradual resorption and exhalation of the gases, which prevents the obstructive bubbles from reforming.

Amniotic Fluid Embolism

Amniotic fluid embolism is the fifth most common cause of maternal mortality worldwide; it accounts for roughly 10% of maternal deaths in the United States. Amniotic fluid embolism is an ominous complication of labor and the immediate postpartum period. Although the incidence is only approximately 2 to 6 in 100,000 deliveries, the mortality rate is up to 60%. Approximately half of those who survive have permanent neurologic dysfunction. The onset is characterized by sudden severe dyspnea, cyanosis, and shock, followed by neurologic impairment ranging from headache to seizures and coma, and by DIC. These features differ from those observed with pulmonary embolism from DVT. Of note, amniotic fluid contains many procoagulants such as activated coagulation factors II, VII, X, and tissue factor. Much of the morbidity and mortality associated with amniotic fluid embolism stems from the activation of coagulation and components of the innate immune system and release of vasoactive substances, rather than the mechanical obstruction of pulmonary vessels by amniotic debris. The vasoactive substances cause acute pulmonary hypertension and right-heart failure, leading to hypoxia, left-heart failure, pulmonary edema, and diffuse alveolar damage.

The underlying cause is the infusion of amniotic fluid or fetal tissue into the maternal circulation via a tear in the placental membranes or rupture of uterine veins. Classic findings at autopsy include the presence of squamous cells shed from fetal skin, lanugo hair, fat from vernix caseosa, and mucin derived from the fetal respiratory or gastrointestinal tract in the maternal pulmonary microvasculature (Fig. 4.17).

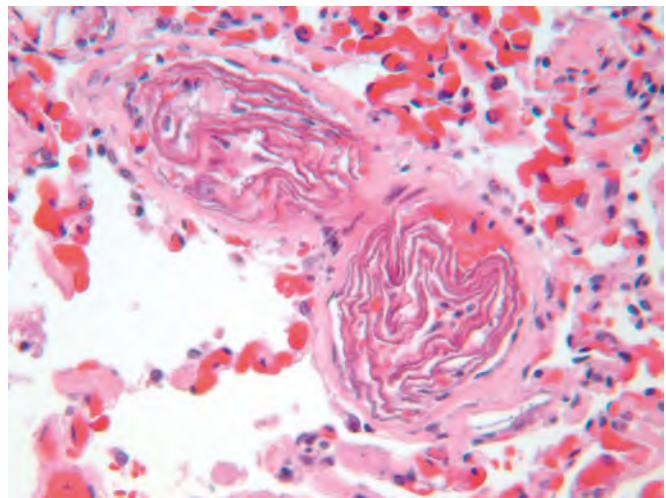


Fig. 4.17 Amniotic fluid embolism. Two small pulmonary arterioles are packed with laminated swirls of fetal squamous cells. There is marked edema and congestion. Elsewhere the lung contained small organizing thrombi consistent with disseminated intravascular coagulation. (Courtesy Dr. Beth Schwartz, Baltimore, MD.)

KEY CONCEPTS

Embolism

- An embolus is a solid, liquid, or gaseous mass carried by the blood to a site distant from its origin; most are dislodged thrombi.
- Pulmonary emboli derive primarily from lower extremity DVT; their effects depend on the size of the embolus and the location in which it lodges. Consequences may include right-sided heart failure, pulmonary hemorrhage, pulmonary infarction, or sudden death.
- Systemic emboli derive from cardiac mural or valvular thrombi, aortic aneurysms, or atherosclerotic plaques; whether an embolus causes tissue infarction depends on the site of embolization and the presence or absence of collateral circulation.
- Fat embolism occurs after crushing injuries to the bones and may cause pulmonary insufficiency and neurologic damage.
- Air embolism occurs upon rapid decompression, most commonly in divers, and results from sudden formation of nitrogen gas bubbles within the vasculature.
- Amniotic fluid embolism is a rare complication of childbirth that is often fatal due to pulmonary and cerebral manifestations.

INFARCTION

An infarct is an area of ischemic necrosis caused by occlusion of either the arterial supply or the venous drainage. Tissue infarction is a common and extremely important cause of clinical illness. Roughly 40% of all deaths in the United States are caused by cardiovascular disease, and most of these are attributable to myocardial or cerebral infarction. Pulmonary infarction is also a common complication in many clinical settings, bowel gangrene is frequently fatal, and ischemic necrosis of the extremities (*gangrene*) is a serious problem in the diabetic population.

Arterial thrombosis or arterial embolism underlies most infarctions. Less common causes of arterial obstruction leading to infarction

include local vasospasm, hemorrhage into an atheromatous plaque, or extrinsic vessel compression (e.g., by tumor). Other uncommon causes of tissue infarction include torsion of a vessel (e.g., in testicular torsion or bowel volvulus), traumatic vascular rupture, or vascular compromise by edema (e.g., *anterior compartment syndrome* of the leg) or by entrapment in a hernia sac. Although venous thrombosis can cause infarction, the more common outcome is just congestion; in this setting, bypass channels rapidly open and permit vascular outflow, which then improves arterial inflow. Infarcts caused by venous thrombosis are thus more likely in organs with a single efferent vein (e.g., testis and ovary).

MORPHOLOGY

Infarcts are classified according to color and the presence or absence of infection; they are either red (hemorrhagic) or white (anemic) and may be septic or bland.

- **Red infarcts** (Fig. 4.18A) occur (1) with venous occlusions (e.g., testicular torsion, Chapter 19), (2) in loose, spongy tissues (e.g., lung) where blood can collect in the infarcted zone, (3) in tissues with dual circulations (e.g., lung and small intestine) that allow blood to flow from an unobstructed parallel supply into a necrotic zone, (4) in tissues previously congested by sluggish venous outflow, and (5) when flow is reestablished to a site of previous arterial occlusion and necrosis (e.g., following angioplasty of an arterial obstruction).
- **White infarcts** (see Fig. 4.18B) occur with arterial occlusions in solid organs with end-arterial circulation (e.g., heart, spleen, and kidney), and where tissue density limits the seepage of blood from adjoining capillary beds into the necrotic area.

Infarcts tend to be wedge-shaped, with the occluded vessel at the apex and the periphery of the organ forming the base (see Fig. 4.18); when the base is a serosal surface, there may be an overlying fibrinous exudate resulting from an acute inflammatory response to mediators released from injured and necrotic cells. Fresh infarcts are poorly defined and slightly hemorrhagic, but over a few days the margins tend to become better defined by a narrow rim of congestion attributable to inflammation. With further passage of time, infarcts resulting

from arterial occlusions in organs without a dual blood supply typically become progressively paler and more sharply defined (see Fig. 4.18B). In comparison, in the lung, hemorrhagic infarcts are the rule (see Fig. 4.18A). Extravasated red cells in hemorrhagic infarcts are phagocytosed by macrophages, which convert heme iron into hemosiderin; small amounts do not grossly impart any appreciable color to the tissue, but extensive hemorrhage can leave a firm, brown hemosiderin-rich residuum.

The dominant histologic characteristic of infarction is **coagulative necrosis** (Chapter 2). Importantly, if the vascular occlusion has occurred shortly (minutes to hours) before the death of the person, histologic changes may be absent; it takes 4 to 12 hours for the dead tissue to show microscopic evidence of necrosis. Acute inflammation is present along the margins of infarcts within a few hours and is usually well defined within 1 to 2 days. Eventually a reparative response begins in the preserved margins (Chapter 3). Most infarcts are ultimately replaced by scar tissue (Fig. 4.19). The brain is an exception to these generalizations, in that central nervous system infarction results in **liquefactive necrosis** (Chapter 2).

Septic infarctions occur when infected cardiac valve vegetations embolize or when microbes seed necrotic tissue. In these cases, the infarct is converted to an abscess, with a correspondingly greater inflammatory response (Chapter 3). The eventual sequence of organization, however, follows the pattern already described.

Factors That Influence Development of an Infarct

A vascular occlusion can cause effects ranging from virtually nothing to tissue dysfunction and necrosis sufficient to result in death. The variables that influence the outcome of vascular occlusion are:

- **Anatomy of the vascular supply.** The availability of an alternative blood supply is the most important determinant of whether vessel occlusion will cause tissue damage. As mentioned earlier, the lungs have a dual pulmonary and bronchial artery blood supply that protects against thromboembolism-induced infarction. Similarly, the liver, with its dual hepatic artery and portal vein circulation, and the hand and forearm, with their dual radial and ulnar arterial supply, are all relatively resistant to infarction. In contrast, renal and splenic circulations are end-arterial, and vascular obstruction generally causes tissue death.
- **Rate of occlusion.** Slowly developing occlusions are less likely to cause infarction because they provide time for development of

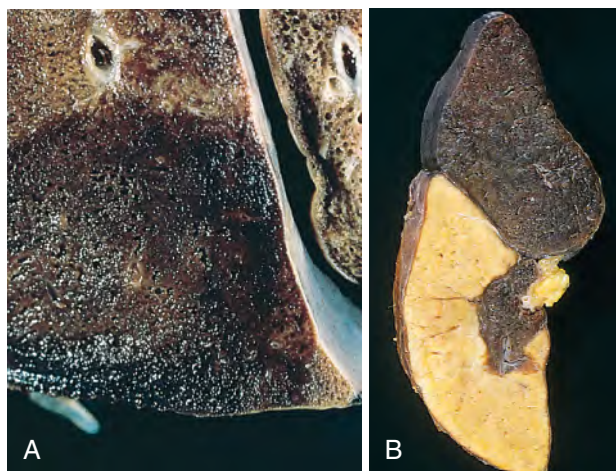


Fig. 4.18 Red and white infarcts. (A) Hemorrhagic, roughly wedge-shaped pulmonary red infarct. (B) Sharply demarcated white infarct in the spleen.

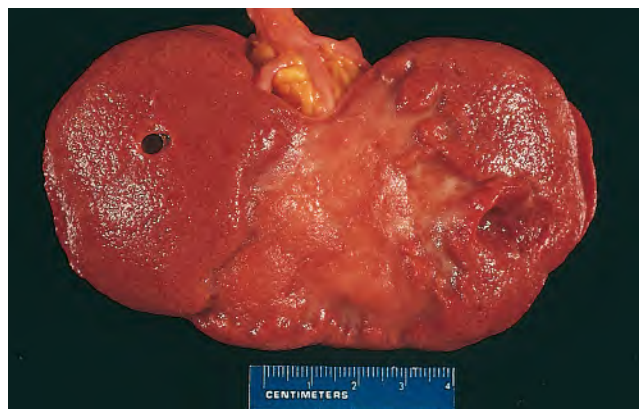


Fig. 4.19 Remote kidney infarct replaced by a large fibrotic scar.

collateral pathways of perfusion. For example, small interarteriolar anastomoses—normally with minimal functional flow—interconnect the three major coronary arteries in the heart. If one of the coronaries is occluded slowly (i.e., by an encroaching atherosclerotic plaque), flow within this collateral circulation may increase sufficiently to prevent infarction, even though the larger coronary artery is eventually occluded.

- **Tissue vulnerability to hypoxia.** Neurons undergo irreversible damage when deprived of their blood supply for only 3 to 4 minutes. Myocardial cells, although hardier than neurons, are also quite sensitive and die after only 20 to 30 minutes of ischemia (although, as mentioned earlier, changes in the appearance of the dead cells take 4 to 12 hours to develop). In contrast, fibroblasts within myocardium remain viable even after many hours of ischemia (Chapter 12).
- **Hypoxemia.** Understandably, abnormally low blood oxygen content (regardless of cause) increases both the likelihood and extent of infarction.

KEY CONCEPTS

Infarction

- Infarcts are areas of ischemic necrosis most commonly caused by arterial occlusion (typically due to thrombosis or embolization); venous outflow obstruction is a less frequent cause.
- Infarcts caused by venous occlusion or occurring in spongy tissues with dual blood supply and where blood can collect typically are hemorrhagic (red); those caused by arterial occlusion in compact tissues typically are pale (white).
- Whether or not vascular occlusion causes tissue infarction is influenced by collateral blood supply, the rate at which an obstruction develops, intrinsic tissue susceptibility to ischemic injury, and blood oxygenation.

SHOCK

Shock is a state of circulatory failure that impairs tissue perfusion and leads to cellular hypoxia. At the outset, the cellular injury is reversible; however, prolonged shock eventually leads to irreversible tissue injury and can be fatal. Shock may complicate severe hemorrhage, extensive trauma or burns, myocardial infarction, pulmonary embolism, and microbial sepsis. Its causes fall into three general categories (Table 4.3):

- **Cardiogenic shock** results from low cardiac output due to myocardial pump failure. This can be due to intrinsic myocardial damage

TABLE 4.3 Three Major Types of Shock

Type of Shock	Clinical Example	Principle Mechanisms
Cardiogenic	Myocardial infarction Ventricular rupture Arrhythmia Cardiac tamponade Pulmonary embolism	Failure of myocardial pump resulting from intrinsic myocardial damage, extrinsic compression, or obstruction to outflow
Hypovolemic	Fluid loss (e.g., hemorrhage, vomiting, diarrhea, burns, or trauma)	Inadequate blood or plasma volume
Septic	Overwhelming microbial infections (bacterial and fungal)	Activation of cytokine cascades; peripheral vasodilation and pooling of blood; endothelial activation/injury; leukocyte-induced damage, disseminated intravascular coagulation

(infarction), ventricular arrhythmias, extrinsic compression (cardiac tamponade; Chapter 11), or outflow obstruction (e.g., pulmonary embolism).

- **Hypovolemic shock** results from low cardiac output due to low blood volume, such as can occur with massive hemorrhage or fluid loss from severe burns.
- **Sepsis and septic shock** are interrelated and somewhat overlapping conditions:
 - **Sepsis** is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection.
 - **Septic shock** is defined as a subset of sepsis in which particularly profound circulatory, cellular, and metabolic abnormalities are associated with a greater risk of mortality than with sepsis alone.
- **Systemic inflammatory response syndrome (SIRS)** is a sepsis-like condition associated with systemic inflammation that may be triggered by a variety of nonmicrobial insults, such as burns, trauma, and/or pancreatitis.
- Less commonly, shock can occur in the setting of a spinal cord injury (*neurogenic shock*), or an IgE-mediated hypersensitivity reaction (*anaphylactic shock*, Chapter 6). In both these forms of shock, acute vasodilation leads to hypotension and tissue hypoperfusion.

Pathogenesis of Septic Shock

The number of cases of sepsis in the United States exceeds 750,000 per year, and the incidence is rising, which is ironically due to improvements in life support for critically ill patients, as well as the growing ranks of immunocompromised hosts (because of chemotherapy, immunosuppression, advanced age, or human immunodeficiency virus infection) and the increasing prevalence of multidrug-resistant organisms in the hospital setting. Despite improvements in care, the mortality rate remains around 40%. Septic shock is most frequently triggered by gram-positive bacterial infections, followed by gram-negative bacteria and fungi. Emerging viral infections such as those caused by SARS-CoV-2 can also cause sepsis.

The ability of diverse microorganisms to cause septic shock is consistent with the idea that a variety of microbial constituents such as *endotoxin*, *staphylococcal enterotoxin B*, and *M protein* of *streptococci* can trigger the process. As mentioned in Chapter 3, macrophages, neutrophils, dendritic cells, endothelial cells, and soluble components of the innate immune system (e.g., complement) recognize and are activated by diverse substances derived from microorganisms. After activation, these cells and factors initiate several complex inflammatory and counter-inflammatory responses that serve to localize and control the infection. Sepsis occurs when the response to infection becomes generalized and injures normal tissues remote from the site of infection. Thus,

sepsis represents the failure of normal control mechanisms that have evolved to contain an infection (Fig. 4.20).

Factors believed to play major roles in the pathophysiology of septic shock include:

- **Inflammatory and counter-inflammatory responses.** In sepsis, various microbial cell wall constituents engage receptors on cells of the innate immune system, triggering proinflammatory responses. Likely initiators of inflammation in sepsis are signaling pathways that lie downstream of Toll-like receptors (TLRs) (Chapter 6), which recognize a host of substances derived from microbes called *pathogen-associated molecular patterns* (PAMPs) and from injured cells called *damage-associated molecular patterns* (DAMPs), as well as G protein–coupled receptors that detect bacterial peptides and C-type lectin receptors such as *Dectins* that detect fungal antigens. Ligation of these receptors leads to increased expression of the genes encoding inflammatory mediators mainly via activation and nuclear translocation of the transcription factor nuclear factor- κ B (NF- κ B). The upregulated mediators include numerous cytokines such as tumor necrosis factor (TNF), interleukin 1 (IL-1), IL-12, IL-18, and interferon- γ (IFN- γ), as well as other inflammatory mediators such as high-mobility group box 1 protein (HMGB1). Markers of acute inflammation such as C-reactive protein (CRP) and procalcitonin are also elevated. Reactive oxygen species and lipid mediators such as prostaglandins and platelet-activating factor (PAF) are also elaborated. These effector molecules induce endothelial cells (and other cell types) to upregulate adhesion molecule expression and further stimulate cytokine and chemokine production. The complement cascade is also activated by microbial components, both directly and through the proteolytic activity of plasmin (Chapter 3), resulting in the production of anaphylatoxins (C3a, C5a), chemotactic fragments (C5a), and opsonins (C3b), all of which contribute to the proinflammatory state. In addition, microbial components can activate coagulation directly through factor XII and indirectly through altered endothelial function (discussed later). The accompanying widespread activation of thrombin may further augment inflammation by triggering protease-activated receptors on inflammatory cells.
- The hyperinflammatory state initiated by sepsis triggers counter-regulatory immunosuppressive mechanisms, which may involve both innate and adaptive immune cells. As a result, septic patients may oscillate between hyperinflammatory and immunosuppressed states during their clinical course. Proposed mechanisms for the immune suppression include a shift from proinflammatory (Th1) to antiinflammatory cytokines (Chapter 6), production of antiinflammatory mediators (e.g., soluble TNF receptor, IL-1 receptor antagonist, and IL-10), lymphocyte apoptosis, the immunosuppressive effects of apoptotic cells, and the induction of cellular anergy.

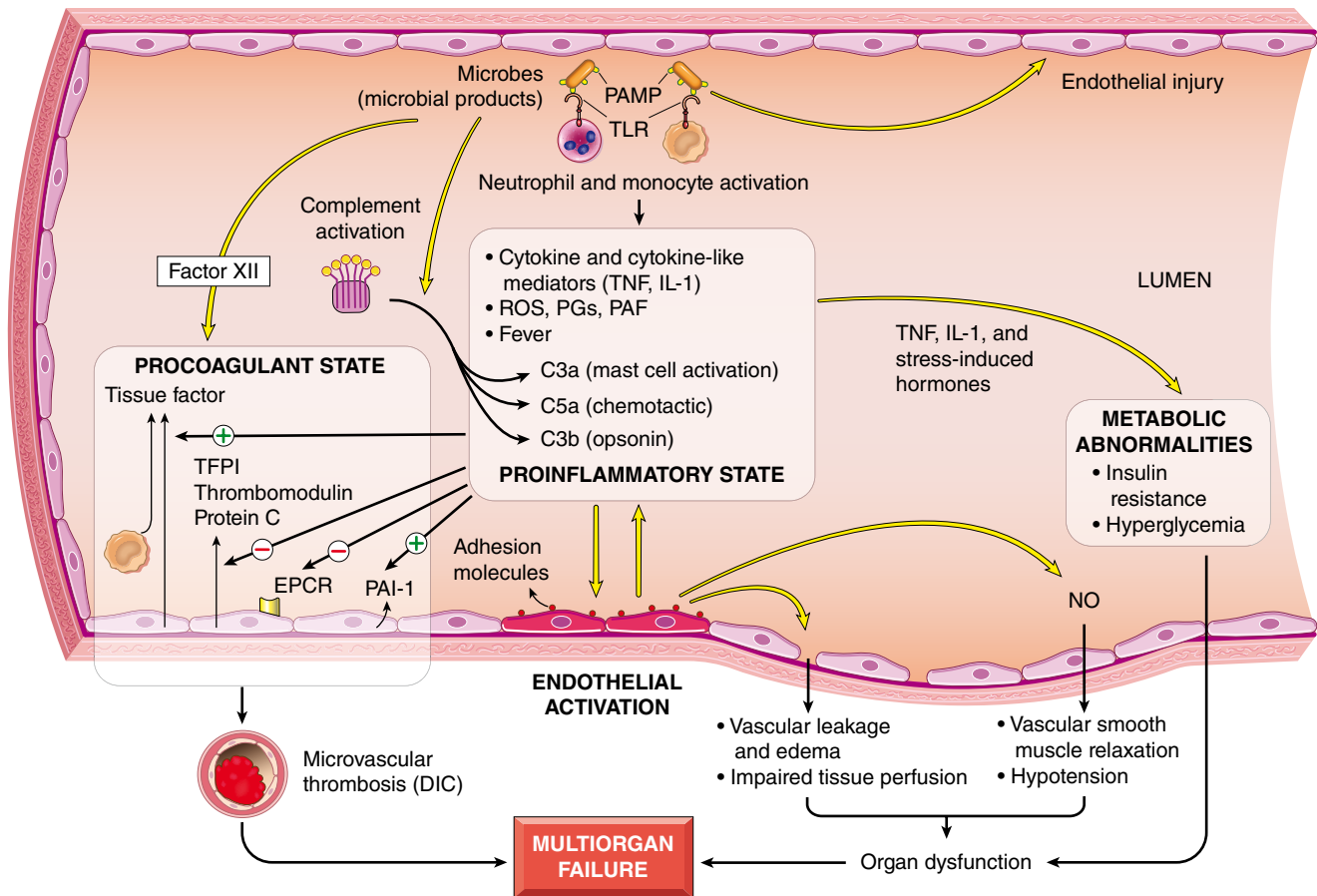


Fig. 4.20 Major pathogenic pathways in septic shock. Microbial products (PAMPs, or pathogen-associated molecular patterns) activate endothelial cells and cellular and humoral elements of the innate immune system, initiating a cascade of events that lead to end-stage multiorgan failure. Additional details are given in the text. C3, Complement component 3; C3a, complement component 3a; DIC, disseminated intravascular coagulation; EPCR, endothelial protein C receptor; IL-1, interleukin; NO, nitric oxide; PAF, platelet-activating factor; PAI-1, plasminogen activator inhibitor 1; PGs, prostaglandins; ROS, reactive oxygen species; sTNFR, soluble tumor necrosis factor receptor; TF, tissue factor; TFPI, tissue factor pathway inhibitor; TLR, toll-like receptor; TNF, tumor necrosis factor.

Current evidence suggests that both pro- and antiinflammatory responses are triggered simultaneously. The intensity of each of these reactions depends on multiple factors intrinsic to the host (e.g., genetics and underlying diseases) and pathogen (e.g., virulence and burden).

- **Endothelial activation and injury.** The proinflammatory state and endothelial cell activation associated with sepsis lead to widespread vascular leakage and tissue edema, which have deleterious effects on both nutrient delivery and waste removal. One effect of inflammatory cytokines is to loosen endothelial cell tight junctions, making vessels leaky and resulting in the accumulation of protein-rich edema fluid throughout the body. This alteration impedes tissue perfusion and may be exacerbated by attempts to support the patient with intravenous fluids. Activated endothelium also upregulates production of NO and other vasoactive inflammatory mediators (e.g., C3a, C5a, and PAF), which may contribute to vascular smooth muscle relaxation and systemic hypotension. Another feature of sepsis is microvascular dysfunction. There is an increase in capillaries with intermittent flow and heterogeneity of flow in various capillary beds, and the normal autoregulation of flow based on tissue metabolic environment is lost. These changes cause a mismatch in oxygen needs and oxygen delivery.

- **Induction of a procoagulant state.** The derangement in coagulation is sufficient to produce the formidable complication of DIC in up to one-half of septic patients. Sepsis alters the expression of many factors to favor coagulation. Proinflammatory cytokines increase tissue factor production by monocytes and possibly endothelial cells as well, and decrease the production of endothelial anticoagulant factors, such as tissue factor pathway inhibitor, thrombomodulin, and protein C (see Fig. 4.10). They also dampen fibrinolysis by increasing PAI-1 expression (see Fig. 4.10). Some studies suggest a role for neutrophil extracellular traps (NETs, Chapter 3) in promoting the procoagulant state by stimulating both intrinsic and extrinsic pathways of coagulation. The vascular leak and tissue edema decrease blood flow at the level of small vessels, producing stasis and diminishing the washout of activated coagulation factors. Acting in concert, these effects lead to systemic activation of thrombin and the deposition of fibrin-rich thrombi in small vessels, often throughout the body, further compromising tissue perfusion. In full-blown DIC, the consumption of coagulation factors and platelets is so great that deficiencies of these factors appear, leading to concomitant bleeding and hemorrhage (Chapter 14).
- **Metabolic abnormalities.** Septic patients exhibit insulin resistance and hyperglycemia. Cytokines such as TNF and IL-1, stress-induced hormones (such as glucagon, growth hormone, and glucocorticoids),

and catecholamines all drive gluconeogenesis. At the same time, the proinflammatory cytokines suppress insulin release while simultaneously promoting insulin resistance in the liver and other tissues, likely by impairing the surface expression of glucose transporter-4 (GLUT-4). Typically, sepsis is associated with a rise in blood levels of glucose, triglycerides, and lactate. Hyperglycemia decreases neutrophil function—thereby suppressing bactericidal activity—and causes increased adhesion molecule expression on endothelial cells. Although sepsis is initially associated with an acute surge in glucocorticoid production, this phase may be followed by adrenal insufficiency and a functional deficit of glucocorticoids. This may stem from depression of the synthetic capacity of intact adrenal glands or frank adrenal necrosis resulting from DIC (*Waterhouse-Friderichsen syndrome*) (Chapter 20). Finally, cellular hypoxia and diminished oxidative phosphorylation lead to increased lactate production and lactic acidosis.

- **Organ dysfunction.** Systemic hypotension, interstitial edema, microvascular dysfunction, and small vessel thrombosis all decrease the delivery of oxygen and nutrients to the tissues that, because of cellular hypoxia, fail to properly use those nutrients that are delivered. Mitochondrial damage resulting from oxidative stress impairs oxygen use. High levels of cytokines and secondary mediators diminish myocardial contractility and cardiac output; increased vascular permeability and endothelial injury can lead to acute *respiratory distress syndrome* (Chapter 13). Ultimately, these factors may conspire to cause failure of multiple organs, particularly the kidneys, liver, lungs, and heart, culminating in death.

The severity and outcome of septic shock are likely dependent on the extent and virulence of the infection, the immune status of the host, the presence of other comorbid conditions, and the pattern and level of mediator production. The multiplicity of factors and the complexity of the interactions that underlie sepsis explain why most attempts to intervene therapeutically with antagonists of specific mediators have not been effective and even had deleterious effects in some cases. As mentioned earlier, another factor in the failure of neutralization of proinflammatory cytokines may be due to the concurrent activation of antiinflammatory mediators, which may tilt the balance toward immune suppression. The standard of care continues to be antibiotics to treat the underlying infection, and intravenous fluids, pressors, and supplemental oxygen to maintain blood pressure and limit tissue hypoxia. Suffice it to say that even in the best of clinical centers, septic shock remains an obstinate clinical challenge.

An additional group of secreted bacterial proteins called *superantigens* also cause a syndrome similar to septic shock (e.g., *toxic shock syndrome*). Superantigens are polyclonal T-lymphocyte activators that induce the release of high levels of cytokines that result in a variety of clinical manifestations, ranging from a diffuse rash to vasodilation, hypotension, shock, and death.

Stages of Shock

Shock is a progressive disorder that leads to death if the underlying problems are not corrected. The exact mechanisms of sepsis-related death are still unclear. Death typically follows the failure of multiple organs. Unless the insult is massive and rapidly lethal (e.g., exsanguination from a ruptured aortic aneurysm), shock tends to evolve through three general (albeit somewhat artificial) stages. These stages have been documented most clearly in hypovolemic shock but are common to other forms as well:

- *An initial nonprogressive stage* during which reflex compensatory mechanisms are activated and vital organ perfusion is maintained

- *A progressive stage* characterized by tissue hypoperfusion and onset of worsening circulatory and metabolic derangement, including acidosis
- *An irreversible stage* in which cellular and tissue injury is so severe that even if the hemodynamic defects are corrected, survival is not possible

In the early nonprogressive phase of shock, various neurohumoral mechanisms help maintain cardiac output and blood pressure. These mechanisms include baroreceptor reflexes, release of catecholamines and antidiuretic hormone, activation of the renin-angiotensin-aldosterone axis, and generalized sympathetic stimulation. The net effect is tachycardia, peripheral vasoconstriction, and renal fluid conservation; cutaneous vasoconstriction causes the characteristic “shocky” skin coolness and pallor (notably, septic shock can initially cause cutaneous vasodilation, so the patient may present with warm, flushed skin). Coronary and cerebral vessels are less sensitive to sympathetic signals and maintain relatively normal caliber, blood flow, and oxygen delivery. Thus, blood is shunted away from the skin to the vital organs such as the heart and the brain.

If the underlying causes are not corrected, shock passes imperceptibly to the progressive phase, which as noted is characterized by widespread tissue hypoxia. In the setting of persistent oxygen deficit, intracellular aerobic respiration is replaced by anaerobic glycolysis with excessive production of lactic acid. The resultant metabolic lactic acidosis lowers the tissue pH, which blunts the vasomotor response; arterioles dilate, and blood begins to pool in the microcirculation. Peripheral pooling not only worsens the cardiac output but also puts endothelial cells at risk for the development of anoxic injury with subsequent DIC. With widespread tissue hypoxia, vital organs are affected and begin to fail.

In the absence of appropriate intervention, or in severe cases, the process eventually enters an irreversible stage. Widespread cell injury is reflected in lysosomal enzyme leakage, further aggravating the shock state. Myocardial contractile function worsens, in part because of increased NO synthesis. The ischemic bowel may allow intestinal flora to enter the circulation, and thus bacteremic shock may be superimposed. Commonly, further progression to renal failure occurs because of ischemic injury of the kidney (Chapter 14), and despite the best therapeutic interventions, the downward spiral culminates in death.

MORPHOLOGY

The cellular and tissue effects of shock are essentially those of hypoxic injury (Chapter 2) and are caused by a combination of **hypoperfusion** and **microvascular thrombosis**. Although any organ can be affected, the brain, heart, kidneys, adrenals, and gastrointestinal tract are most involved. **Fibrin thrombi** can form in any tissue but typically are most readily visualized in kidney glomeruli. **Adrenal cortical cell lipid depletion** is akin to that seen in all forms of stress and reflects increased use of stored lipids for steroid synthesis. Whereas the lungs are resistant to hypoxic injury in hypovolemic shock occurring after hemorrhage, sepsis or trauma can precipitate diffuse alveolar damage (Chapter 13), leading to so-called “**shock lung**.” Except for neuronal and cardiomyocyte loss, affected tissues can recover completely if the patient survives.

Clinical Features

The clinical manifestations of shock depend on the precipitating insult. In hypovolemic and cardiogenic shock, patients exhibit hypotension, a weak rapid pulse, tachypnea, and cool, clammy, cyanotic skin. As already noted, in septic shock, the skin may be warm and flushed owing to peripheral vasodilation. The primary threat to life is the underlying initiating event (e.g., myocardial infarction, severe hemorrhage, bacterial infection). However, the cardiac, cerebral, and pulmonary changes

rapidly aggravate the situation. If patients survive the initial insult, worsening renal function can provoke a phase dominated by progressive oliguria, acidosis, and electrolyte imbalances.

Prognosis varies with the origin of shock and its duration. Thus, more than 90% of young, otherwise healthy patients with hypovolemic shock survive with appropriate management; by comparison, septic or cardiogenic shock is associated with substantially poorer outcomes, even with state-of-the-art care.



KEY CONCEPTS

Shock

- Shock is defined as a state of systemic tissue hypoperfusion resulting from reduced cardiac output and/or reduced effective circulating blood volume.
- The major types of shock are cardiogenic (e.g., myocardial infarction), hypovolemic (e.g., blood loss), and septic (e.g., infections).
- Shock of any form can lead to hypoxic tissue injury if not corrected.
- Septic shock is caused by a dysregulated host response to bacterial or fungal infections; it is characterized by endothelial cell activation, vasodilation, edema, DIC, metabolic derangements, and organ failure.

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